VISTA
ARCHITECTURE ROADMAP HEADLINES (INTERNAL ONLY)

for the ♥ of cinema

Version history

Version	Date	Who	Comments
0.1	22/03/2022	John Hanson	Copied into format for external consumption
0.2	25/03/2022	John Hanson	Added context information from larger doc & review
0.3	28/03/2022	John Hanson	First External Distribution
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Overview

Ultimately, an Architecture Roadmap is an outcome of three different things:

- The current state of the software, its issues, and its shortcomings.
- The organizational and product outcomes that the business would like to achieve.
- How technology solutions can facilitate these outcomes.

To be clear, whilst this document must be closely aligned to the long-term product roadmap (and touches some of these items), this document is aimed squarely at the engineering underpinnings of our software, not product features.

This document does not explore Vista Digital or Security in depth. A small section at the end is included for interest, but this section should/could not be seen as exhaustive.

It should also be noted that this document is a summary document of a much longer Vista Architecture Roadmap document, so some many details are omitted for (somewhat more) brevity.

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Desired Architectural & Business Outcomes

Vista Group Purpose

Our purpose is to bring more people together to experience the magic of movies and cinema by creating the platform that connects the industry and powers the moviegoer experience.

Vista Group Objectives

Three key objectives are outlined:

- Re-establish customer trust, ensure we are fit for growth.
- Deliver our core platform to measurably improve our customers businesses (Vista Cloud and EQ)
- Increase market share and seek to maximize the opportunities from our portfolio

Vista Desired Technology Outcome (or "Why did we write this document?")

Key to objective two is the adoption and success of Vista Cloud.

As we transition from a Product Company to a Service Provider, we need to evolve the technology platform to be fit for purpose. The primary goal of this Vista Architecture Roadmap is to enable us to reach a mature architectural state where all Vista customers can be onboarded and supported in Vista Cloud.

This is at the heart of this document, and whilst there are other key outcomes below that are important to the ongoing success of Vista, this document is primarily focused on how we can adapt our platform for success in Vista Cloud.

We Create and Maintain a Mature Cloud platform

The core of Vista's technology journey is to move to a world where we can host and deliver our product suite to customers on the cloud at scale.

The benefits are numerous, but include:

- It ensures customers can benefit immediately from the waves of innovation delivered by Vista. This will increase stickiness and enable our teams to deliver real business value to every customer as soon as it's ready to do so.
- Having our customers (and their data) in the cloud will accelerate the ability of other Group companies to make sales and win more customers, as the integration is easier, and undoubtedly enable us to develop new offerings not currently in the market.
- It will ensure we are able to attract and retain the best people – people who want to work on leading edge technology.
- It will generate incremental revenue opportunities – as Vista is able to charge for more 'layers' of the solution (and for services that we don't yet offer which are enabled by a SaaS delivery approach).
- It will allow us to modernise our older applications, keeping us ahead of competitors in this market.
- Customers will benefit from the power of cloud scalability, meaning that ticket sales can always be available, no matter the load.
- It lowers the barrier to entry for existing customers to use more Vista Software through add on capabilities.

To try and understand how products should mature from their current state to a mature SaaS state, it's worth documenting the elements that make a SaaS Solution mature, and ultimately the destination we want to move all our software to over time. They can be described as follows:

Multi Tenanted

In a mature SaaS product, all elements of the software and hardware stack will be shared across all Tenants. Whilst the ideal end state for the majority of Vista's software stack (and our goal), there may be some elements that that would not

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suit some of Vista's product suite due to a variety of reasons, including isolation of customisation workloads as well as dependencies in third party software libraries that have no equivalents outside of Windows/Full Framework.

For those parts of Vista unsuitable to be multi-tenanted, a deployment stamp based Architecture is a logical end state for many of our products and services. The deployment stamp may be for a region, to reduce the cost of latency sensitive operations for customers, or down to the individual Tenant basis to separate execution and extensions. Please see the Multi-Tenancy section further in this document for more information.

Single Version

A Mature Service will not allow a customer to choose what version of software they reside on, nor to stay on a version for an extended period.

The only exception to a 'single-version' rule is with Blue/Green deployments where updates are being rolled through the system progressively meaning that two versions exist for a temporary period.

To accomplish this, products will need to mature in several ways. These include:

Test Automation Coverage

To trust rapid, broad deployment of releases into production environments, test automation coverage must be significant, representative of real customer data and scenarios, and cover all core key product flows.

This also includes performance and security testing, as services that are functionally correct, but fail at load or expose security vulnerabilities are as useless as services that have functional issues.

Version Tolerant

Whilst a single version of a given service will run in the cloud, components that talk to it could be spread across a set of versions. As such, backwards compatibility will need to be factored into decision making as rolling out new versions across the estate.

Configured, not Customised

With the execution of a single version for a customer, we are no longer able to create components that replace default components to differentiate execution behaviour.

Instead, such components and behaviours need to be flagged in configuration so that these behaviours only occur for a single/group of customers. In some cases, these behaviours need to be isolated to mitigate systemic risk of cascading failure during execution taking down the host process.

Feature Flagged

In an on-premises world, upon getting a new product version, customers would generally have unlimited access to all functions. In the world of SaaS, we can separately charge for sub functions within a product, increasing revenue and our ability to monetise operations that may be only required by a subset of customers.

We also need to track this usage and activation for billing purposes on a per tenant, periodic basis. It also allows us to release code and features that are lower risk, as they will only be enabled for a subset of customers.

Upgrades are Progressive

In Vista's world, not all customers are the same, and there is a lot of variation between customers, and potentially sites. Thus, it's important that we enable software rollouts through the following mechanisms:

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1) Environment Rings – The ability to release software to Test/UAT environments before production environments.

2) Region Rings – The ability to release software into one geographic region before others.

3) Tenant Rings – The idea that we can push traffic for one or more Tenants to new software before others.

4) Release Rings- The idea that we can push traffic for one or more Sites to a new version within a Tenant before others.

Not all mechanisms may be utilised during an upgrade, but these mechanisms can be used to make the release of software a safer process. It is important to note that these rings are purely for risk mitigation purposes; a single version will roll through all rings and eventually be deployed for all customers.

Can Roll Back

It is unavoidable that bugs will be shipped to production, due to unforeseen outcomes, manual mistakes, incomplete test coverage, configuration complexity etc. Thus, whilst we would love to be able to fix a mistake and ship a fix to production immediately, there are likely to be situations where we may need multiple hours/days to resolve an issue. Thus, it is essential that we can revert the state of a system to a previously identified good state quickly without causing further issues.

Use of Open-Source best of breed technologies

Open-source technologies are favoured over in house created solutions to reduce ongoing development costs and increase our ability to use additional open-source plugins and libraries to further extend our functionality. It also makes it easier to on board new staff who are much more likely to have experience with these open-source tools that they are with the bespoke Vista equivalents.

Closed-source, commercialised, best-of-breed technologies should also be evaluated on a TCO basis in preference to in-house creation of commoditised capabilities.

Commercialisable under different models

This allows us to modify our billing models to fit the deal that we may be making. From a fixed yearly value through to a per member or per ticket model, it allows customers and our sales teams the right strategy to come on board.

Vista no longer licenses “products”, rather licenses capability that has elements that reside in a set of existing products.

Scalable

Perhaps an obvious point, but our offerings need to be able to scale with new customers coming on board as well as the major demand spikes that come with blockbuster releases or promotions.

Easy and Consistent Deployment

With Full SaaS, we are no longer beholden to periodic releases and slow upgrade cycles. It is important therefore that we move away from older deployment schedules and methods into new systems.

This includes:

- Fast Release Cadence: To allow fast release of features and fixes into production.
- Blue/Green and Canary: To allow releases to be staged incrementally to give confidence in the upgrade process.

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- Micro Releases: Rather than having months' worth of work released, smaller, incremental releases allow us to add business value quickly and validate fixes for correctness and for business impact. At which point it is a business decision which features are held back into larger groupings for other reasons (marketing/customer appetite)
- Configuration as a Service: It's important that we are no longer beholden to configuration systems that are difficult to modify or understand. Configuration has often been spread across multiple systems and duplicated on a per application basis.
- Roll Forward: In general, fixes will be rolled forward where possible, but it is inevitable that we will need to roll back releases.
- Build Process: Builds will be done through scalable systems that allow test automation and deployment straight into test and production environments seamlessly.

Adhere to Cloud Architectural Principles

Since Vista will be hosting, deploying, and monitoring the systems, we are no longer constrained by lowest common denominator, on-premise systems. Rather, we can pick deployment platforms that fit our business goals the best.

Hence, we aim for the following:

- Immutable Servers/Containers: Servers need to be treated as cattle and new servers/containers are to be deployed rather than upgrading existing components in place. This gives us a fixed deployment, allowing for us to scale and deploy safely.
- Fully Automated: No servers or containers should have manual elements in their creation or maintenance. Manual elements create potential inconsistency in results, due to configuration drift, and costs in manpower.
- Containerised: Ideally, all components will run in Linux containers (or in specific circumstances, Windows Containers) to isolate execution and allow for best practice modern operational practices whilst retaining portability between cloud providers (see next point).
- Cloud Agnostic: We very likely need to deploy across multiple clouds, especially with China. Thus, we should not be beholden to Cloud specific implementations, but can leverage them if there are functional equivalents on other providers.

Provisionable & Manageable

There are likely to be some elements that are only for technical staff to manage (such as an upgrade), however, it is very likely that a multitude of additional roles and business functions will need to interact with the software. Therefore, it is important that we expose tooling appropriate to those teams such that they can use the system.

This could be to extract billing information, stage a new customisation such as a payment connector, or license a new module.

Business Value is Visible

It is tempting to think of SaaS services as a set of web services and a database. Even if the product is operational and transactional in nature and not designed for end users, business stakeholders from the customer should be able to see visual artefacts of the system that give them confidence in its operation and its ongoing value to their business.

Observable

These products and environments are (by definition) complicated and have a lot of moving parts. If we require customers to notify us of issues (performance, security or functionality) then we ultimately are not adding value to on premise offerings.

Thus, observability must be baked into these products to succeed.

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Thus, we need to:

- Have indicators for business activity health (transactions in process/loyalty members/throughput) etc
- Have indicators for individual service and endpoint health, including response times and response codes
- Allow for alerting rules to make sure that people don't need to be constantly looking at dashboards and log files, rather they are pro-actively notified of issues to look at.
- Implement monitoring and alerting systems that include built-in anomaly detection to avoid constant 'whack-a-mole' manual creation of alerts in response to outages

Supportable

After an incident has been identified (hopefully pro-actively and early) it's important that we're able to resolve that issue and get the product back to full health.
Thus we need a good suite of diagnostic tools available to enable us to get to the heart of the issue quickly.

Efficient

One of the key reasons we are moving forward with SaaS is to create solutions that can be delivered to customers economically. Much of this can be achieved by having products Multi Tenanted, however if each tenant uses significant resources, the aggregate total may be uneconomical.

Thus, it is important that small customers brought on have near 0 incremental cost and the overall cost of the underlying resources used are maintained as a small percentage of the overall service price. Examples of such savings may be the reduction in SQL resources due to creating good indexes, removing CPU bottlenecks in app code or switching to alternate state providers if transient data needs to be stored.

Performance testing will be used to detect and reject any linear correlation between tenants and resource usage.

Secure

Products and services are now, by definition, public. Thus, they will come under much more scrutiny and attack than those of on-premises systems of specific customers.

Thus, we need to secure these systems at a high level. This includes:

- Embedding secure development practices into our processes.
- Have good user authentication and authorisation. Ideally externalised to vendors
- Protection against attack (such as DDOS and targeted attacks).
- Auditable for logins and sensitive operations.
- Not vulnerable to common attack vectors such as SQL Injection and XSS.

Compliant

With the move to the cloud, Vista is now in scope for our customers compliance requirements to external and government stakeholders, so we will need to develop and release in accordance with appropriate compliance standards. PCI have introduced new standards that come into effect in 2022 that we must comply with. For other Vista software it is unclear at the present time exactly which standards we're going to adhere to, but it's likely that at minimum it will include SOC2 Type 2. Technology and supplier decisions must be made with compliance in mind.

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Resilient

To give our customers confidence in the platform, we'll need to supply an SLA to them for services. What that SLA will look like will depend on the services purposes, but it is important that we treat potential SLA breach events extremely seriously given the reputational and financial impact they may have.

The service can be adopted incrementally

It is very likely that a customer's journey to the cloud will not be 100% overnight. Rather, it is much more likely that customers will move their operations incrementally. Thus, communication channels and version assumptions cannot assume that the rest of a customer's estate will be on the cloud, or on Vista's cloud. Whilst Vista can make some assumptions about models that will be supported by Vista, and versions that customers must run on premise to stay current, there will be some element of interoperation for a while.

Migration is achievable

Customers have a large amount of data in on premise environments and integrations and extensions to existing systems that are critical to their business operation.

To attract these customers to SaaS, we need to allow customers to migrate their data and extensions onto the cloud, in a timely and safe fashion. This is often known as providing 'safe-passage' to the SaaS service.

Migration is Easy

We wish to bring on as many Vista Cloud customers as possible, as quickly as possible, in a managed and supportable manner. The platform must enable the fast onboarding of new Tenants and Sites, whilst reducing the risk that such a migration will create operational issues in future.

We create and maintain a compelling Vista Digital Platform and Product Suite

Vista's current Digital Channels have been developed in parallel, over time using a variety of technologies. This has created several issues for us and for our customers. Amongst a great many more, these include:

1) A single enhancement needs to be done in four separate places. Web, Kiosk, iOS and Android. This makes features expensive and slower to release to market.
2) Due to (1), our sales channels have differed in functionality due to time or cost constraints, with some functionality only existing on a single platform.
3) The configuration of features is different on each channel, with a single feature needing to know how to set up three different types of platform with three different settings to enable a single feature.
4) Whist the user interfaces have been created to tailor to the form factor involved, the varying ages and toolchains involved have meant that users have no consistency in their experience with the platform.

Omnichannel Clients

Our goal for digital is to leverage the OCC components (and APIs) that have been created over the last two years to create a single, unified sales channel platform, with the transaction (and user interface) logic created in a single place and then leveraged on each individual platform separately. We use these components in concert with services such as CXM, MovieXChange Film CDN, Umbraco CMS and others to create a compelling suite of products that power the next

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generation of Cinema sales channels that differentiate themselves from the competition and empower organisations such as Movio to manifest their product right into the sales process.

It should be noted that each platform will have components and capabilities bespoke to them, such as interacting with payment hardware for Kiosk and camera access on the mobile app, but the core logic will be shared.

Vista Digital Platform

Is the next generation of Vista Digital API's, facilitating both the OCC created Sales Channels as well as a multitude of other third-party solutions to deliver outstanding performance, scalability, security and availability.

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We enable Data Driven Decision Making

Much of the basis of making a good decision is the reliance on the following key attributes:

a) Good quality data to make the decision on.

b) Mechanisms by which stakeholders can search for answers to their questions.

c) Mechanisms to allow for systems and algorithms to notify people that relevant data suggests an action should be taken.

Vista and our customers can end up making the wrong decision in a multitude of areas given that we have not captured sufficient information of good quality and do not have mechanisms by which to then make the right decision at the right time.

This manifests in a multitude of ways:

- In Vista's own Product Planning.
- In how Vista creates and tests its software.
- In how Vista effectively react to issues encountered in the field.
- In how our customers can predict/adapt to market demand.
- In how quickly our customers can react to events.

Thus, it must be a focus of Vista's to capture much more data, in better ways and then allow for more robust and comprehensive analysis methods to allow for good decisions to be made in an expedient manner.

Ideally, as we mature, the true value of this data is our ability to start to use this data in automated ways to notify users that an action is recommended and/or act without human intervention automatically to create better outcomes for our customers.

We Amplify Vista Group and External Partnerships

Vista's success over the years has been driven by its wide range of products and solutions to meet the needs of exhibitors around the world.

In addition, this success has been driven by the extensibility of our product; with hundreds of integrations created by customers, third parties and Vista Group partners to extend the functionality to cover functionality that Vista itself has not.

One of the core elements of Vista's SaaS vision is that the platform will enable a Marketplace where Vista's core Software can empower a wide range of participants the ability to enhance the Vista product offering to our customers and be compensated for it.

Thus, as Vista's solution moves to be SaaS delivered, these integrations (and future ones) create additional challenges that do not currently exist on premise. These include:

- Licensing and Access control – To make sure that only authorised integrations can interact with customers they should have access to.
- Resource and Bandwidth Usage – Now that a customer could be on the cloud, ill-behaving (or just frequently called) integrations could incur significant server and bandwidth costs for Vista.
- Upgrade Compatibility – With an increased cadence of upgrades, not only do we have to be concerned with how Vista's software interacts with our own product, thus, we need to make sure that upgrades are not going to affect third party integrations.

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- Diagnostics – Traditionally, issues with on premise integrations could be diagnosed directly by the customer/integrator though tools available to the exhibitor such as log files and request tracing. Many of these tools will no longer be directly accessible to our customers, placing this support burden on us.

Thus, it is important that we enhance our integration methods to consider these new elements, moving away from a model where every new integration is a bespoke development in a specified format to a world where large parts of our functionality can be exposed via well supported and licensed API's.

We can Attract, Retain and Empower Top Quality Staff

A primary asset of Vista is the quality and domain knowledge of its staff. No matter the discipline, Vista has been lucky enough to attract and retain an amazing set of staff members who contribute their significant amount of experience and passion to helping create compelling software for Vista, Cinemas and Cinema Patrons around the world.

It is vital for the success of Vista and the success of its strategic goals that we continue to attract and retain great quality staff and give them the tools upon which they can continue to create great software.

Examples of this could be:

- Reduce barriers to creating and releasing software (e.g. the migration to more modern and efficient source code and build systems, and the reduction of the number of supported versions of software over time).
- The migration of legacy software to newer platforms or languages.
- Enhancing deployment and configuration systems to decrease the time to test a software change.

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Current State of Vista Cloud & Digital

Vista Managed Servies "Cloud V1"

Vista had been transitioning over a few years to unlock the potential of running its software in the cloud, but that process had been somewhat slow as previous conversations with customers showed a neutral (at best) willingness to move their IT systems to the cloud.

This changed in 2017 with our customers becoming more willing to entertain a move to the cloud, and the business began funding a specific project to add cloud support to our products. When Vista added Cloud capability to its software for Vista Managed Services it essentially meant two things.

Cloud support for cinema sites

The software that has traditionally run on an application server at each site can now be centralised to a commercial cloud vendor. That meant that services, tasks, configuration, and application data are relocated from each site to a set of resources in a central location. These are then shared across multiple sites, reducing the total number of servers required. A circuit that previously had to run a hundred servers to run their estate could now run on a dozen.

This decision drove a lot of change in our software. This included (but is not limited to):

- Changes to all SQL in Vista to support databases and Cinema Servers configured to run in the UTC timezone.
- The ability for any service to run workloads for potentially hundreds of sites (rather than one).
- A completely new Task Scheduler and Executor system capable of running hundreds, or thousands of tasks simultaneously across a number of servers.
- New security mechanism for registering clients and securing services.
- The ability to run Back Office on a single machine for dozens of users, working over different sites.
- The ability to run more than one version of Vista software concurrently on the same set of Cinema Servers, supporting the progressive rollout of a new version across the sites in a circuit.
- Changes to support centralized logging from all services, including the ability to track log entries across multiple servers on multiple machines.
- A concerted effort to move our Back Office Management application to the web (namely, to Cinema Manager).

To be blunt, this was the largest set of platform changes that Vista had gone through in over a decade, and was in large part the precursor to all subsequent changes.

Commercial cloud support for Vista central products

Second, exhibitors can now make good use of services offered by commercial cloud providers in a number of our products. The major change was that 'SQL as a service' offerings such as SQL Azure and AWS RDS for SQL were added as supported database platforms for Vista Cinema, VGC, Loyalty and Vista Connect.

Whilst smaller than the changes required for Cinema, this nonetheless enabled a lot of efficiency gains in our central product suite.

More details around the changes required/made can be found here.

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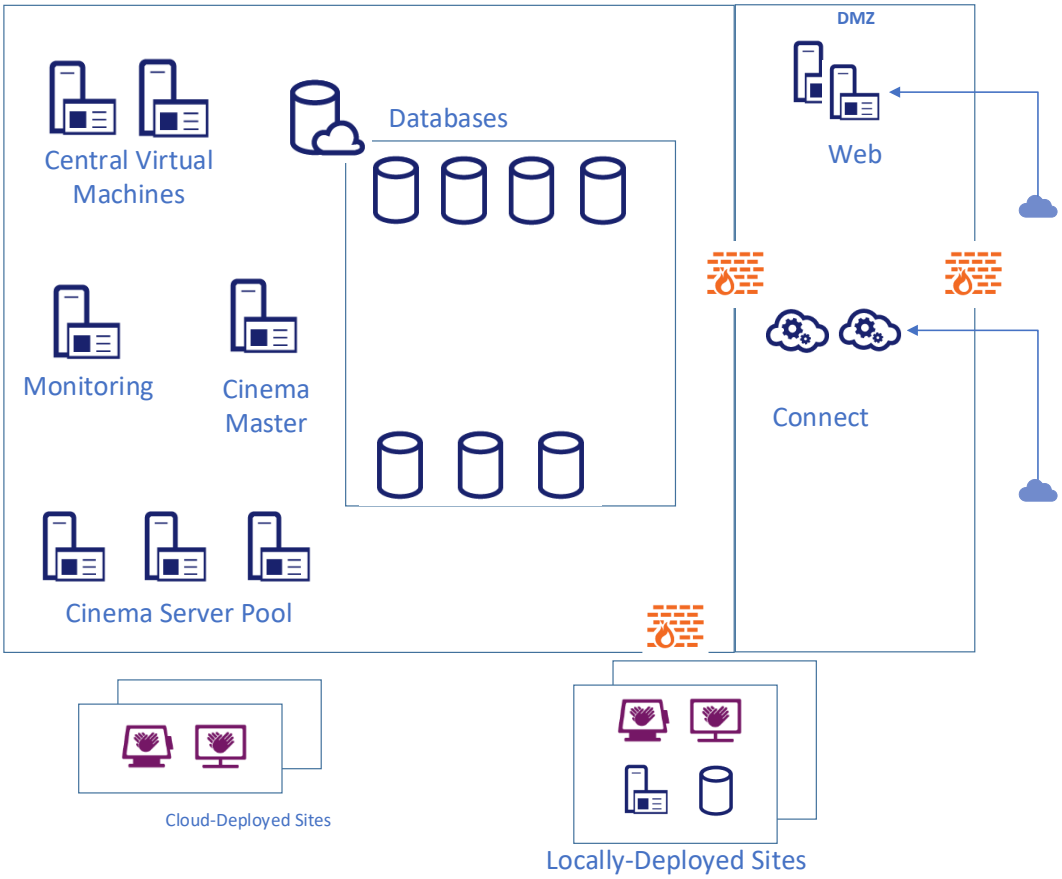
This diagram below shows a high level architectural overview of the resultant product suite as seen in a typical Vista Managed Service deployment:



This is the product that Pathé has used for several years in France, and more recently in Netherlands and Belgium, for their entire circuits. Further, a deployment of Vista's central modules in Azure has served Odeon well since their roll out, as their central elements are hosted and managed by our outstanding Cloud Operations and Services Teams.

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Vista Cloud

The current generation of Vista Cloud software used by Wallis is built directly on the underpinnings of Vista's Managed Service offering.

As we looked at the market uptake of Vista Managed Services, and what we'd like to be part of Vista Cloud, it was clear there were several functional and platform gaps that needed to be filled.

A more compelling User Experience

We wanted Vista Cloud to first and foremost feel like a coherent user experience for users. This meant that we had to do a serious amount of work to get from a model that felt very much to users like "Vista on Premise, with Servers further away" to a feel more like a single coherent SaaS platform.

Highlights of this included:

- A new Central Identity Management System that enables single sign on by a user, using their enterprise authentication details such as Azure AD, to access all the products that are available to them (rather than being forced to use dozens of separate logins on different technologies). Much more detail can be found here.
- A greatly increased web footprint for Vista where Head Office, Loyalty, Vouchers and Giftcards, and Cinema. All are entirely navigable through a single web interface. The user can run reports, switch systems and sites, and look at all the available data entities without noticeably cutting away from this web interface. More details around the underlying Aspect library can be found here.
- A new Form Launcher product that extends the single web navigation experience to include desktop forms that are yet to be converted to the web. A single click in the web navigation pane downloads, executes and logs user into the right place to continue their work without having to log into and navigate an entirely disconnected session.
- A new landing page enabled all users key circuit performance data at their fingertips, enabling them to get key numbers from any device anywhere.
- The integration of Vista Horizon directly into the platform enabled further insights to be gathered and explored from the same navigation plane as everywhere else. More detail can be found here.

More Administrable

Hand in hand with the operational enhancements for end users, it was important that we gave some IT administrators for customers additional tools to improve their experiences and win them over, including the following:

- The new Application Launcher system enables very quick download and installation of client application such as POS and Kiosk. This was combined with our new Central Identity Management System into our Client devices, allowing a manager to "register" their workstation on Vista cloud with a QR code scan on their phone, enabling secure access to all the functionality that that Client can provide as securely as possible.
- Central Identity Management's integration into externally federated identity providers enables a single point at which administrators can onboard, and importantly offboard staff to all system access. This also enables administrators to set their own 2FA rules (such as SMS validation code access). As a result, CIM enables customers compliance efforts and eases governance concerns.
- The Central Identity Management System User Interface for IT Staff allows the setup of coherent and consistent user rights across the circuit, enabling access to a half dozen systems for a user in seconds.

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- The integration of Task Scheduler's new Web UI enables access to configuration of, and visibility of the entire circuit's IT task execution form the same Landing Page as other functions are accessed.

More Scalable

Vista's underlying deployment mechanism changed with Vista Cloud, now deploying Servers into "Scale Sets", enabling a customer to move from a pair of servers in capability to potentially dozens if required for a blockbuster event with a single mouse click, or a single rule applied to monitor current system utilisation.

Hard won lessons from the limits of scale in the Pathé system also made their way into Vista cloud, enabling us to make much more efficient use of the resources available to execute workloads.

More Monitorable

A large amount of effort was put into improving our ability to identify and diagnose issues proactively and retrospectively compared to the previous iteration of Vista Managed Services. This included

- The ability to view the status of POS devices throughout the customer estate, including if they are offline from the internet unexpectedly.
- Additional visibility and logging of several areas, with work on logging formats making sure that many more areas were feeding our central logging systems correctly.
- The introduction of a mature Application Performance Monitoring Suite in the form of AppDynamics, a market leader in that space.

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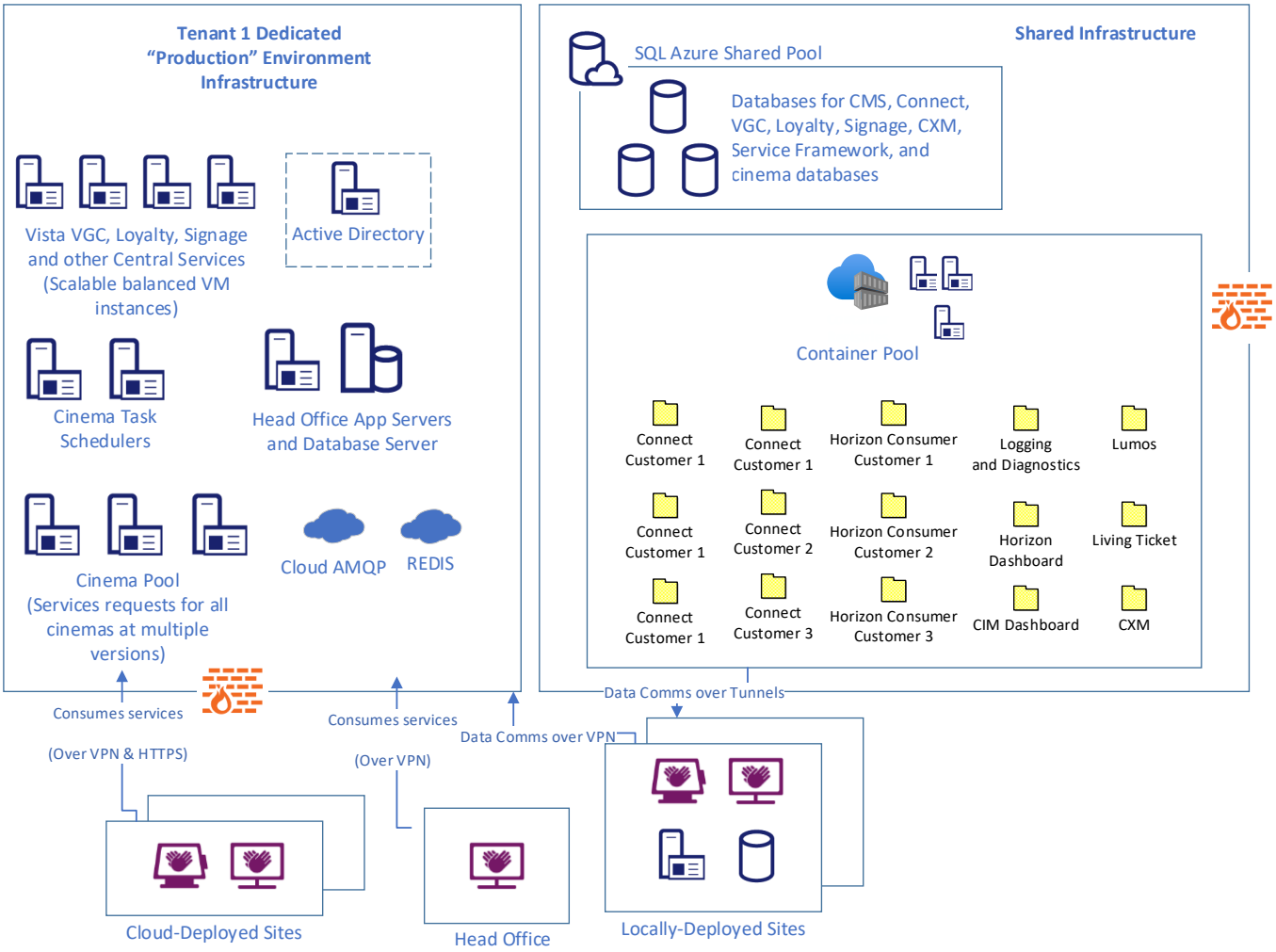
High Level Architecture Diagram



Detailed Architecture

Can be found here if you would like more details around the current Vista Cloud Architecture.

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Vista Digital Platform

In addition to the Vista Cloud Operational Suite, an incredible amount of work went into our Vista Digital Platform product. Traditionally called Connect on premise, the platform underwent a great deal of engineering effort to move to the cloud and act as a much more "Cloud Native" product.

A small selection of highlights includes

- The removal of all Virtual Machine usage, moving to containers (and Cron jobs for tasks), where we can share infrastructure between Tenants, scale more quickly, and take advantage of off-the-shelf operations tooling.
- These containers, whilst Windows Containers, were split by application service enabling the independent scaling of highly used parts of the Digital Platform (over others).
- The addition of connectivity through the new "On-Premises Proxy/Circuit Gateway" service, allowing for secure, VPN free access to all the servers and services we need access to on premise with a simple Vista add-in upgrade. In addition, all direct DB access was removed from the Web Loader. More detail can be found here.
- A great deal of additional test automation cases was added to help give confidence in our ability to upgrade more quickly.

In addition, a great deal of time has been spent to create the business functionality required for our next generation digital offerings through the maturing of the Omni Channel API. A new API suite tailored to these channels, and only licensable in future through the Vista Digital Platform, it is what underpins our Omni Channel Components, now and into the future.

Details around many of the Architectural elements can be found here.

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Vista Digital Channels

While we've been creating the next generation of our Operational Software suite, a similar transformation has been occurring in the Vista Digital space.

Traditionally, our customer facing channels have been created through the hard work of a number of teams in Vista in entirely disconnected code bases, often talking to entirely different back-end systems that would need to be enhanced, installed and configured separately to get a cohesive result across channels.

This meant that single functional update to add a feature would need change in:

- Vista Sales Server
- Vista Connect
- Vista Kiosk
- Vista Web
- Vista Mobile iOS
- Vista Mobile Android

Combine this with Call Centre, POS, Serve, Rove and Group Sales Public bookings and MovieXChange, and you quickly found that a single innovative feature was spectacularly hard to do, and just as hard for the customer to get their hands on as it required wholesale upgrades of multiple systems.

Vista Omni Channel Components

A solution to reduce this barrier to innovation was envisioned in the form of Vista's Omni Channel Components. A set of JavaScript (/Typescript) components that could perform all the actions required to create an entire Sales Channel from end to end, created in a way that you could mix and match, and style components together to deliver the right user experience for the right place in the channel.

Vista OmniChannel EcoSystem

Several other shared building blocks have been created to underpin our new Sales Channel offerings that further enhance the offering in market. These include:

Vista Headless CMS

The other major building block for our new Sales Channels, we have used a commercial CMS solution (in the form of Umbraco) to create a fully multi-tenanted linux-hosted platform that allows the customisation of all their sales channels in a single place. This includes layout, colours, fonts, text and images, enabling a compelling, consistent (omni-channel) experience across all their public facing channels.

Living Ticket

Powered in part by our Horizon technology, it enables the creation of a compelling e-ticket experience for patrons. A ticket that can contain custom advertising, moving elements as well as the ability for the ticket to be updated in near real time with any changes that occur at cinema. This includes seat or auditorium swaps.

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The living ticket is also a mechanism that allows us to join together many parts of a customer's interaction with a cinema, from the original sale, through to the pickup of an order, the purchase of a drink, and their in-seat dining options to give far more visibility into how customers are interacting with a Cinema and the total value of their patronage.

MovieXChange Film

This is a set of services and software for us to deliver consistent, rich multimedia experiences for all our channels. This includes the automatic management and update of Film Images and Trailers, as well as meta data about the film's actors, directors, runtime and many more. MXF CDN was added as a global-scale content distribution service to deliver media assets to end-users at any scale, and is currently used by multiple customers including for Odeon's web and mobile channels, for example.

Customer Experience Manager (CXM)

This creates a core personalisation ability for our Sales Channels, enabling the ability for us to customise experiences that best fit individual patrons.

This includes the ability to use film propensity data generated by Movio to be utilised to generate individualised film displays and recommendations within the Sales Channels.

As time goes on, the goal is to enable Vista and Vista Group companies such as Movio to use CXM to alter many more elements of the sales experience, giving a great competitive advantage over competing (non-cinema specific) competitors. Examples include the ability to give specific deals, incentivising upsells or modifying behaviour (for example a switch to a premium showing).

Vista Digital Sales Channels

Omnia and Lumos

To be clear, there are two broad categories of Vista Digital Sales Channel.

- Lumos: An out of the box, configurable solution that is lower cost and gives customers access to a great deal of Vista Digital Functionality.
- Omnia: Offered from our Vista Digital agency, they use the OCC building blocks as well as custom development to create a bespoke set of digital channels that match the exact requirements and vision of the customer.

Due to the bespoke nature of Omnia, the rest of this section refers to Lumos channels.

Vista Lumos Web

Our new Web offering, an attractive website that combines all the above elements to deliver a customer website that enables and easy and compelling way to buy tickets, concessions, sign up for loyalty programs and more.

Created on a multi-tenanted Linux base, with Cloudflare at the front, it's designed to be both cost effective for us to run and very fast for the customer (no matter where they are in the world) to use.

We have also created a cut down version of this as an F&B microsite, to enable customers with existing websites they're happy with (or are not yet ready to move over) for in seat dining and pre-show concession ordering.

An example can be found here.

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Vista Lumos Mobile

A singular mobile platform built with the same technology building blocks as web (and with a couple more such as Ionic), to enable the same great user experience and functionality offered on web, but through a much more native feeling mobile tailored app.

An example build can be found in the Apple App Store here.

Vista Lumos Kiosk

The currently final piece of the Vista Digital puzzle, it gives a compelling in-cinema Kiosk product that is built upon and makes use of all the above Vista Digital building blocks.

Running in a Windows Shell, it enables us to use all the existing Windows Integrations (such as Barcode Scanners/Card Readers/Pinpads/Printers et al), but with a much richer customisable experience that fits with the overall digital suite.

An Android Kiosk is also possible, with the exact same functionality, but with differing hardware integrations, further delivering on the Omni-Channel Dream.

Current Issues & Barriers to Growth

Vista Digital Platform

The core software in the Vista Digital platform is Connect, a piece of software that's been around 20 odd years and has had, for most of its lifespan, assumed a very specific operating environment and connection environment.

A great deal of work has gone into transitioning this product into a cheaper to run, more maintainable and quicker to deploy solution. However, significant work still needs to happen to transition the platform to a place where it can be deployed to every Vista customer in the world and form the basis of ticketing in the Cinema Industry globally.

Vista Digital

At present, the Vista digital channels are still in their relative infancy, and whilst they are built correctly, with the long term in mind, a lot of continued investment is required to meet the market and smooth the product edges as it evolves. This includes product elements such as making Kiosk software work on Android, through to more technical platform elements like Server-Side Rendering of code to reduce page load times and improve SEO performance.

There is also the not insignificant matter of migration from existing 'classic' Vista Web, Kiosk and Mobile to their Vista Digital replacements – currently this looks like a manual process and is, in effect, a rebuild from the ground up. This approach, while clean, is unlikely to scale to where we need it.

Vista Cloud

The preceding section hopefully gives a good feel for just how far the product and platform has been driven over the last 5 years, from a solely on-premises product that is installed off of installation packages by customers on physical machines located at Cinema, to a world where Vista is hosting customers in a central location, with a product that looks and feels at home as a SaaS offering. It's worth celebrating that achievement as a monumental shift from what we were doing a few short years ago, much of it done through a global pandemic that impacted Vista and our customers greatly.

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However, the work to get to this point is likely dwarfed by the work that's required to get the system to the business outcomes we wish to deliver. These include:

- The ability to onboard, manage and upgrade every current Vista customer (and yet more to come) on a single platform in an efficient and cost-effective manner.
    - At present, a great deal of manual effort is required for every customer and every environment to do any one of these steps. This is bad for a handful of customers but is unmanageable at scale.
    - As a result of this world wide uptake we will need to meet the needs of organizations in a plethora of countries . Much of Vista's 20 years has been spent creating custom integrations and regional extensions for a wide range of customers and countries. The vast majority of which were built with no consideration for Cloud concerns. For example, Fiscal Systems were created and predicated on constant DB availability (as it was attached via a LAN cable). This assumption is no longer true, and would otherwise mean for a great deal of customers that they could not trade offline in any manner. There are hundreds of such integrations spread across dozens of integration types.
    - The ability to scale the platform to meet the requirements of ticket sales for future blockbusters for all these customers. Whilst we can scale at present, it is not yet cost effective or fast enough.
- We want Vista cloud to be a truly compelling market offering, continuously delivering innovation at pace to meet the demands of an ever-changing exhibition market.
    - At present, we must stage feature releases periodically (4 or more months) to avoid inevitable bugs with changes to our incredibly complex software base, and corresponding UAT overhead and live outage outcomes. We need to do all we can to allow us to develop, test and release value to our customers quickly, safe, repeatably, and predictably. Releases need to be boring and trivial.
    - We also want to unleash the power of Vista Group's resources, along with other participants in the Cinema Software industry to be a force multiplier to our own efforts. At present, we don't have a good way to enable these technical and commercial relationships and opportunities to be actioned.
- We want to be the trusted partner for all cinema exhibitors no matter their size. Trust manifests in a lot of ways but includes:
    - What you can demonstrate on paper (through certifications like SOC-2 and PCI)
    - On screen (such as dashboards, performance metrics and Audit Records)
    - Through our daily actions (like pro-active incident monitoring)
    - What happens when something inevitably goes wrong (like POS offline capability, system patching and security incident response), how long it lasts and how much it impacts their business.

We need to do this all progressively, over time, improving both the platform and service regularly and incrementally to build to a state where we think we can honestly say we've delivered all the above.

The devil's in the many, many details outlined in the subsequent sections of the document, but the headline is that, from a SaaS platform perspective, it is not yet mature and has a long way to go before it can fulfill the business requirements that we have set for the platform.

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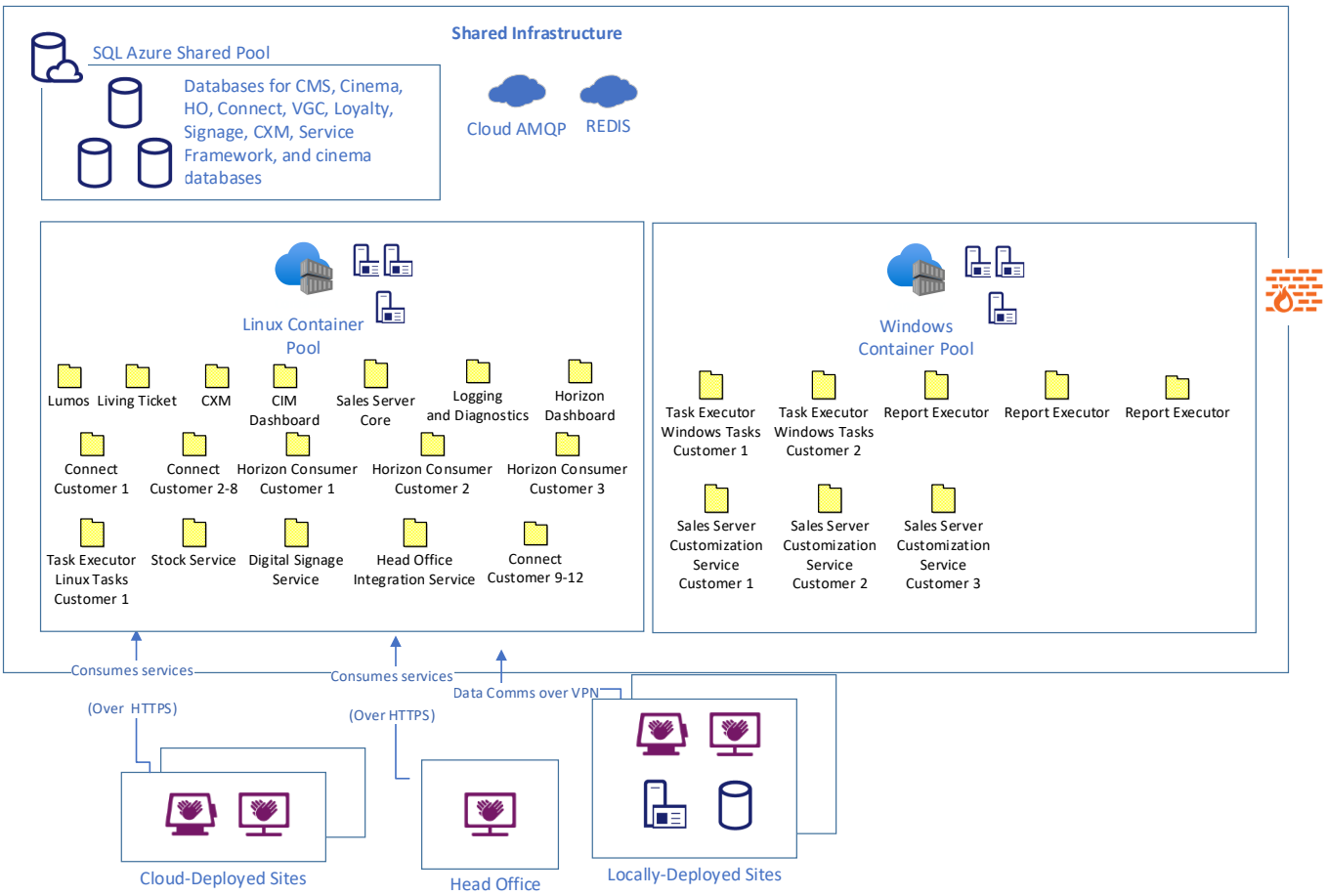
Architectural End State Diagram and Summary



Architectural Building Blocks

After discussing the current state, it's worth summarising (at a very high level) the end state we'd like to get to. Whilst it should be noted that this diagram is not exhaustive, it contains many key examples of the end state Architecture we'd like to migrate to. These include:

Multi-Tenanted Linux Containers

These make up most workloads in Vista Cloud. Whilst Multi-Tenancy is a key goal, it should be noted that the Connect Container image above may be split in reality:

- Some customers (such as Customer 1) may be Isolated due to a requirement for Private Cloud (i.e. Fully Isolated Workloads), or via our own choices in how best to manage the intersection of risk, cost and performance. Very large customers may be isolated for transaction critical workloads to reduce risk or reduce the impact of large cache refresh operations.
- Some customers (such as Customer 2-8) will be grouped together to indicate that they constitute a "release ring" in a single region. This is distinct from Customer 9-12, which are isolated into a ring that could move earlier-later than that related to Customers 2-8.

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- It should be noted that we could/should be able to have customers in different release rings on a per product/service basis (depending on the relevant risk factors), with some customers potentially being in an earlier release ring

Single Tenant Linux Containers

In some cases, we may choose a deployment stamp architecture for our Linux Containers, to allow for workload isolation. Either to reduce workload risk, or for components that would be too difficult to multi-tenant but that can be moved to Linux. It's likely that this group will be small, as multi-Tenancy is preferred to Linux Containerisation as it relates to workload migration.

Multi-Tenant Windows Containers

A possible migration step along the way to maturity for some products, some services may well stay as Multi-Tenanted Windows Containers. Such an example may be Report Execution, where we need access to Windows only Crystal Execution binaries, but we are less concerned about isolation of workloads.

This is likely the most common deployment model after Multi-Tenanted Linux Containers.

Single Tenanted Windows Containers

Another possible migration step for some components, it is likely that several services may remain in this state in perpetuity. This may be for a variety of reasons, but most likely because:
a) There is a barrier to converting to Linux. Such as third-party dependencies on Windows, or a long tail of dependencies/systems that would need migration.

And
b) There is a barrier to multi-tenancy. This is most likely due to the risks of having multiple customer workloads impacting others in the same space.

A good example of this Architecture is the Sales Server Customisation Service, where wildly different workload mixes and binaries will be required to run. Separation of these extensions allows us to reduce cross tenant execution risks, both with noisy neighbours, as well as dependent binary differences.

Task Execution

System tasks, wherever possible will be migrated into a Multi-Tenanted Linux execution environment, allowing for as much density and scalability as possible. These are the tasks that are going to be executed frequently and thus benefit the most from this level of optimisation.

However, most tasks have been created by Vista's remote offices for specific cases/environments and often connect to/access Tenant specific resources. As a result, the Vista Cloud Tasks Scheduler will be extended to allow execution of tasks across all environments, but the execution container will be Windows based and scoped to that specific Tenant (as per the previous section). This gives us a fall back for very bespoke or difficult to convert items, that allows us to concentrate on higher value conversions/work items.

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VPN Usage

It is likely that the cost of removing VPNs for hybrid deployment scenarios (such as having On-Premises sites) may not be viable compared to the benefits. With that one specific scenario excepted, it is imagined that all external traffic will not need a VPN for operation.

Databases

It should be noted that at present, it is not envisioned that existing Vista Databases will be multi-Tenanted. Multiple decades of effort have gone into creating databases with very specific scopes and purposes. To change this would be an epic undertaking, which could consume years of effort with very little pay off compared to the capital and opportunity costs of such an effort. This would also make migration more difficult, forcing us to create migration tooling to move customer data from one schema to the next.

The existence and support of Azure Elastic Database Pools and Managed Instances gives us flexibility at a database level to manage workloads as we see appropriate to consider both the growth of Compute as well as Storage in the pools. Azure's upcoming Hyperscale Elastic Pools may give further density and scalability again.

It should be noted that whilst we are not tied directly to Azure through this decision (as SQL Server can run on a multitude of other clouds, including those run by our China team), there is a financial incentive for us to Stay on Azure due to the maturity of their PaaS offerings in this space.

One potential saving grace here is for services that may have some workloads migrated away only in the Cloud for specific purposes. It is likely that we would further migrate these workloads away from SQL server entirely (for example temporary order management in Sales Server), but we could reasonably consider Multi-Tenanted SQL instances as a landing place if we have workloads that fit well into an RDMS use case. If enough high impact workloads are moved off the standard databases, then the cloud portability story is more viable financially.

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Delivery Phase Outcomes

Introduction

These phases pull together the individual line-item phases below and summarise the business value at each, hopefully giving the platform multiple step change evolutions in capability, saleability and supportability through our journey to maturity.

We're looking for continued ways to open up new markets, enable more customers to join Vista Cloud and complete the required pre-requisites for future phases.

Phase 1

Overall Goal

Refine the current Vista Cloud solution to expand our capability to onboard and manage more customers, giving room for work that takes the platform forward significantly for future phases.

New Market Opportunities

- Add the ability for Loyalty and VGC to be included in Vista Digital Platform. Initial cut is like VMS/Vista Cloud Architecture, and thus may be limited in viability to customers that have 10+ sites.
    - Many Vista Digital Customers will not need these additional products, and thus this more limited platform will still be viable for smaller customers.
- The first API is offered externally through a platform that will continue as the API Marketplace, beginning a journey of creating a community around Vista Cloud and a simpler on-boarding onramp.

Enabling more Customers to join Vista Cloud and improve the Vista Cloud Experience

- Reduce costs for Vista Digital Platform and increase deployment velocity (of both new customers as well as new versions)
- Minimise human effort required for creation and upgrades of customer environments.
- Improve our visibility of customer environments and alert on more failure conditions.
    - Including App/Forms Launcher as well as Task Execution
- Reduce impact of Software updates on Customer WAN environments, especially for POS updates.
- Some critical POS offline improvements give confidence to an additional group of customers that Vista Cloud is safe for their workloads.
- Extend CIM to make it more usable and useful
    - Extend the number of integrated products
    - Allow customers to have lower cost users for some users
    - Reduce onboarding effort from existing systems
    - Allow customers to on-board staff before their start dates
    - Enable Regional Fault resilience
- Improve our ability to update customers, reduce fault scenarios and enable rollback if issues are found with an update.
- Progress on Certification gives confidence for larger customers to bring their workloads to Vista Cloud.

Setting up for Future Delivery Phases

- Begin Migration of core components to .Net Core to enable future Linux Containerisation.

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Phase 2

Overall Goal

Significant effort begins on moving the VDP platform forward, whilst delivering Vista Cloud improvements in critical areas and setting up major platform re-engineering in Phase 3 & 4.

New Market Opportunities

- VDP can now act as a stepping-stone to Vista Cloud. Loyalty and VGC are re-engineered along with Connect to be much more cost efficient and Cloud Native. These are the vanguard for future elements.
- Capabilities are now saleable, allowing for additional revenue and upsell opportunities.

Enabling more Customers to join Vista Cloud and improve the Vista Cloud Experience

- We have brought down CTS enough to address customers that have 5+ sites.
    - A choice could be made to further decrease that number if we were to run a singular VM that contains Cinema/HO management products with a decreased (internal or external) SLA.
    - A further choice is open to us to use Windows Containers as an interim step for elements not yet brought to a more mature architectural state.
- Vista Digital CTS is now engineered such that It's viable to support customers with single sites for Loyalty, VGC, VDP , Omnia & Lumos offerings.
- Some critical POS offline improvements give confidence to an additional group of customers that Vista Cloud is safe for their workloads. Critically, this includes multi-POS Seat Allocation whilst offline.
- Additional observability further increases confidence in the platform.
- Many key Vista Cloud workloads are moved to Containers to reduce overall Vista Cloud CTS.
- Onboarding enhancements to reduce the load on project teams to bring a customer to Vista Cloud or Vista Digital Platform.
- Releases are more stable, performant and secure with additional test automation.
- The Vista Marketplace has matured to the point where several integrators are using this as their primary connection mechanism and new customers are using this as a mechanism to transition on-premises workloads.
- VGC and Loyalty can be used in a VDP context without a VPN, further reducing barriers to adoption.

Setting up for Future Delivery Phases

- Enable future Linux Containerised, Multi-Tenanted Components through shared network and service infrastructure. This will be the standard for the majority of Vista Software.
- Improve our ability to create and execute automated test environments (and by extension Tests themselves).
- The introduction of a new function based framework decreases development and delivery time for new bespoke features allowing us to ship better code faster.

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Phase 3

Overall Goal

More product features can be updated regularly, demonstrating the true power of SaaS software for customers and futher differentiating the experience from on premise. It is faster and cheaper to onboard customers through the further refinement of migration tooling and the cost to serve has dropped through major operational and infrastructural savings.

New Market Opportunities

- Offline support, especially for Fiscalisation/Ticket Numbering and Receipt Numbering opens additional market opportunities in several new regions, including Central Europe and South America.

Enabling more Customers to join Vista Cloud and improve the Vista Cloud Experience

- Containerisation/Multi Tenancy of transactional elements enable multi-data centre failover if databases are also replicated and enabling a compelling performance profile compared to on-premises and an ability to scale in/out at high speed.
- VPNs are removed for clients at site enabling simpler 4G/5G backup links from the Cinema and/or device. 4G roaming now a possibility.
- Most services now are separated in source, deployment, and hosting. Enabling the structure of teams/support to support a DevOps model.
- Additional observability further increases confidence in the platform. Importantly, this also includes visibility by customers.
- Releases are more stable, performant and secure with additional test automation.
- The Vista Marketplace has matured, and more integrators are using this as their primary connection mechanism. Including to on-premises sites.
- CTS greatly reduced due to targeted reduction in usage in specific App and DB tier areas.
- Customers now have visibility into Vista Cloud operations, enabling them to get visibility (and gain trust) of the platform.
- We’re now set up to deliver features incrementally (rather than 3 times a year) for many product areas.
- Customers have far more access to audit records and reports enabling our largest customers to satisfy more internal and external audit requirements.
- Onboarding tooling increasingly matures to enable increased velocity of customers coming onto Vista Cloud.

Setting up for Future Delivery Phases

- Large section of our codebase, along with key infrastructural building blocks are now able to be run on Linux. This enables us to begin mass migration of workloads to Linux Containers.

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Phase 4

Overall Goal

- Significant cost reductions, scalability & stability improvements are released into market, enabling us to take on additional customers of varying sizes (including much smaller customers). This is especially true as it relates to the workload elements of Vista Cloud that have non-periodic spikes, such as blockbuster sale events.

Enabling more Customers to join Vista Cloud and improve the Vista Cloud Experience

- Vista has enabled a DevOps support model for the vast majority of the product suite.
- Additional observability further increases confidence in the platform.
- Releases are more stable, performant and secure with additional test automation.
- The Vista Marketplace is now a recognised integration point for our teams, third parties and customers. This includes
    - The full removal of VPN's (when everything is in the Cloud) creates the first real feeling of SaaS Cloud for Vista.
    - Customers have far more access to audit records and reports enabling our largest customers to satisfy more internal and external audit requirements.
    - Customers can gain access through the marketplace to more capabilities, created by Vista and by third parties.

Setting up for Future Delivery Phases

- We now have all the primary building blocks we need to continue converting all Vista software and refining current products.

Phase 5

Overall Goal

- Vista Cloud is now a trusted platform by a great many customers that has proven to be stable as well as deliver continual innovation.

New Market Opportunities

- Offline support may be able to be enabled through the release of a Managed appliance, satisfying late adopters/cloud sceptics of the viability of Vista Cloud. (Not a certainty at present)

Enabling more Customers to join Vista Cloud and improve the Vista Cloud Experience

- Crystal Reports has been replaced with Horizon reporting.
- Head Office uploads are deprecated, with the supported data pipeline being the Horizon one.
- Head Office Downloads undergo major refactoring, enabling a greatly improved master data experience.
- No Cabinets forms remain, completing one of the last vestiges of the feel of On-Premises software.
- Releases are more stable, performant and secure with additional test automation.
- VPNs are removed from all situations and deployment topologies.
- No VM workloads remain.
- Most services run in Multi-Tenanted Linux Containers

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Roadmap Delivery Headlines and Phasing

The remainder of this document breaks down the architectural roadmap into broad thematic areas, describing the following properties for each theme.

- Title – The theme title and a numeric reference to the Architecture Roadmap document which contains far more detail and is recommended reading for further context.
- Problem statement – An overview of the reason we need to address this theme, with some examples of the concrete problems we are facing in this area
- Solution – A high-level description of the proposed solution, coupled with examples of defined goals and outcomes that we are hoping to achieve.

The document then breaks deliverables within each theme into a phased delivery approach, providing an indication of how we will take Vista through this multi-year journey of work.

Each work item should help deliver in one (or likely more than one) of the following areas:

1) Adoption: How quickly and smoothly can we onboard customers and then maintain them in perpetuity?
2) Trust: Can we build trust in the platform, firstly to enable the customer to feel comfortable in a move to the Cloud, then to stay there?
3) Cost: How can we continue to bring down the costs of serve, both operationally and infrastructurally?
4) Innovation: How can we deliver great software that demonstrates the true value of Vista Cloud and get it into our customers hands quicky?

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.net 6+ Conversion (7.2)

Problem

At present, most Vista software is created using the Microsoft ".net Full Framework." A Windows only framework, the last version of which, 4.8, was released in 2019. No new features will be added to this version, and tool and library support is fast moving away from the platform.

The platform will approach the status that VB6 currently holds over the next few years, and it will become increasingly difficult to support the platform as well as find/retain technical staff to work on it.

Microsoft's replacement is ".net Core", now confusingly named ".net 6.0". This version is faster than .net 4.8, better supported and gives us the ability to move Vista Cloud to run under Linux, a major cost saving as well as a much better supported platform in terms of operational capabilities.

In addition, some limited Vista software remains in VB6, a platform last released in the early 2000's and something impossible to get to run in most Cloud scenarios.

Solution

Two key deliverables are envisioned through the process:

- We would like all Vista Server Components to run on .net Core on Linux. Most software can be moved from ".net Full Framework" to ".net core" without change, but there are a number of areas where Microsoft has changed/deprecated libraries or requires additional change for support on Linux. This allows us to retain the great deal of business logic we've created over the last 20+ years with lower risk onto a much more viable long-term platform for Vista.

At the end of this process, we'd like 90% of Service workloads to have moved from Windows to Linux running on .net 6+, with the only remaining software on ".net Full Framework" being the software that for good reasons (whether incompatibility, or high cost and low return) is not chosen be converted.

- VB6 is entirely removed and is not used anywhere in Vista Cloud Server environments (excepting some limited POS/Kiosk usage under exceptional circumstances)

Note, that during the various updates to software, during the conversion process we will also likely be adding other best practice changes as practical to the software to minimize the number of regression test cycles that are applied. That includes

- Converting the code to c# (where it is likely to be commonly modified by developers in future)
- Adding Unit/Integration Tests
- Adding static code analysis to catch common errors and ensure good coding practices
- Adding Monitoring hooks and confirming logging is appropriate
- Adding "Cloud Mode" features to make sure that Multi-Cinema and UTC operations are enabled
- Enabling Linux Support (for example removing Registry requirements)
- Changing the Source Control/Build Outputs for Install Customization to move away from a mono-repo mindset and enable much more deployment flexibility.

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Phases

Phase 1

- A project to convert our shared libraries to .NET Standard 2.0 will be undertaken. The libraries prioritized for delivery will be those used by software currently being hosted under Connect, Loyalty, VGC and Core Service Framework but should also continue through common infrastructural elements.
- Service Framework will have a net 6.0 template to work from to allow new services to be created from. On Premises Deployments will enable .net 6.0 Services via Install Customization.

Phase 2

- Shared Library's progress to .net Standard 2.0, with a goal to deliver this phase's goals, as well as preparing for the next phase.
- Loyalty and VGC and associated components are converted to .net 6.0.
    - A Loyalty sidecar may need to be created for Loyalty Plugins, but at present it's imagined that this will not be needed, and we would convert these plugins in place. If unviable, then a sidecar pattern service may need to sit alongside.
- Integration Components (such as Payment Connectors and Fiscal Drivers) can use .NET Standard 2.0 where possible. This may necessitate changes in Sales Server as well as Client Software such as POS where these components are shared with Sales Server.
- Legacy and Cloud Task Scheduler will be modified to support .NET Standard 2.0 Tasks.
- Task Scheduler Executors will allow for execution of both Windows and Linux nodes.
- New Services, Integrations and Tasks, wherever possible will be created to run under ASP.NET Core and .NET 6.0 (or .net Standard 2.0 where a plugin). This likely means that templates need to be updated.
- Core Service Framework Services and low hanging fruit extension services are converted to .net 6+.
- Task Scheduler will be converted to .net 6.0
- All Connect Services and Tasks will be converted to .net 6.0.
- High Volume Transactional Services previously worked on in Thunderbird will be released as .net 6.0.

Phase 3

- Additional IIS and Service Framework and elements changed to run in .NET 6+.
- Tasks (where possible) will be updated to .NET Standard and create .NET Standard NuGet packages, to be run under a Linux based executor.
- Commonly supported integrations/plugins will be updated to .NET Standard.
- VB6 components that are still run on a Server or in Forms Launcher will be updated to .net 6+ (or Full Framework if required)
- Sales Server will be split into constituent parts and the Core elements moved to .net 6+
    - This likely means that extensibility points for Sales Server will need to be separated into a Full Framework only service that runs as a Sidecar to Sales Server.

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Phase 4

- All Services will be updated to run on .NET 10+
- The balance of less common, but still supported, integrations will be updated to .NET Standard wherever possible.

Phase 5

- Remaining Vista Components, including Client Applications are converted to .net 6+ if technically feasible.

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Multi-Tenancy (7.3)

Problem

The current Vista Cloud Architecture is based around the concept that each individual Customer and customer environment (for example a Test Lab vs Production) has a running set of software (and by servers/server resources) dedicated to that customer. Traditionally called a "deployment stamp" this architecture closely mirrors the equivalent concepts which have underpinned our on-premises software for 20 years.

Thus, whilst a pragmatic option to bridge our on-premises software to the Cloud it has several challenges that are insurmountable as we scale to increasing numbers of customers and sites, becoming exponentially more difficult over time.

These issues include

1) Even if we upgraded all our software to run in a containerized workload, which was thus able to share underlying infrastructure, there is a significant cost to keeping that specific customer/environment running, even without a single ticket being sold.
2) By creating a system where each customer may have two more sets of software always running to cater for sites being progressively updated to a new version, it creates a few hundred environments all running while need to be monitored, managed, upgraded and looked after. Any form of continuous deployment of new features becomes virtually impossible due to the overhead involved in update processes.

   As we move from Virtual Machines to Linux containers, this problem becomes worse, as the number of objects moves into the thousands, the failure of any one of which could bring down a customer's operations. There are simply too many spinning parts to look after and places where something could go wrong.
3) This architecture makes it difficult to set up a robust Disaster Recovery strategy for customers, as the creation of a separated Disaster Recovery region for a customer forces a duplication of cost.

Solution

Vista's ideal application architecture end state is that of sharing as much computational capacity as possible. The most efficient way to do this is via the multi-Tenanting of services at the application tier, allowing for workloads to be shared across all customers in a given region (and potentially those who are not usually located in that region, but have had to be migrated for disaster recovery purposes).

This is how Veezi currently operates, as do many of the services written for Vista Cloud (such as CIM & DAS). We would eventually like to see 95% of our software running in this model to rationalize both infrastructural and operational spends down to a reasonable (and profitable level).

We do see a couple of areas where multi-Tenancy would be very difficult to achieve, with significant investment into areas that potentially only benefit a single small customer. Thus, our architecture, will be somewhat of a hybrid, with all core workloads running through multi-Tenanted services, and with extended operations (for example, a highly-specific integration with a niche member provider in South Africa) running under a single tenanted hosting model, allowing us to reduce the number of "per-tenant" pieces of software down to a much more manageable number.

Ultimately, the incremental cost to onboarding two sites for a new customer should be similar in nature to that of onboarding two sites for an existing customer, both in operational and infrastructural expenditure.

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Phases

Phase 1

- No progress. Some design/thinking to go on.

Phase 2

- Shared Library & Infrastructure Support
  - Database Connection Retrieval, Logging and Metrics
- Service Framework Core Services will be multi-Tenanted and released as such.
- Task Executor Services will be modified to allow multi-Tenancy.
  - A single tenanted execution mode will still be enabled at this step due to the lack of conversion of all tasks.
- Cloud Task Scheduler will be converted to allow scheduling of Central Tasks, with Multi-Tenanted Executor (Linux). Windows execution will run single tenanted for the foreseeable future as it enables a low touch option for legacy tasks. The Scheduler itself may remain multi-Tenanted in Linux depending on the risk/rewards/cost of conversion.
- Loyalty & associated Services will be multi-Tenanted.
- VGC & associated Services will be multi-Tenanted.
- Core tasks will be updated to enable for multi-Tenanted execution.
- Release Mechanisms for Steering traffic to appropriate regions/service Tenant-Rings.
- Work will be undertaken to enable compatibility with Veezi Multi-Tenancy models, allowing for Vista and Veezi workloads to be shared where practicable.
- Connect will be Multi Tenanted.

Phase 3

- Remaining Transactional Services are multi-Tenanted
- Most Tasks can run in Multi-Tenanted Systems. (Linux Executor)
- Most Plugins can run in Multi-Tenant Systems.
- Sales Server Core will be multi-Tenanted.
- Core Transaction Services will be multi-Tenanted.

Phase 4

- Continue Multi-Tenanting Elements as practicable including Service Framework Services.
- All Service Framework Queue Processors are multi-Tenanted/put into new framework
- Most IIS services are multi-Tenanted.

Phase 5

- Completion of all Multi-Tenancy that is practicable.

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Containerisation (7.4)

Problem

Due to our beforementioned deployment stamp architecture (and the fact that we use full Virtual Machines for deployment), we have a cost of multiple thousands of dollars per customer that is onboarded.

Deployments are often slow and risky, as each Virtual Machine represents dozens of services, each containing a different functional area, developed, and tested by dozens of teams.

Scaling can also be slow and inefficient, with a whole set of Services being duplicated to add capacity in one very specific area.

Solution

We would like all Vista Server Components, (wherever technically possible) on Linux Containers. Most of these containers should run fully Multi Tenanted at the application level, with some containers running the scope of the Tenant where the security, technology or operational requirements mean that this is not financially viable (possibly as Windows Containers). No VM based solutions remain unless there are literally no other options available to us.

Phases

Phase 1

No progress

Phase 2

Task Executor will be dual hosted, once as a Multi-Tenanted Linux Container, and once as a Single Tenanted Windows Container. The goal is to reduce the number of Tenants that need the latter provisioned but giving the option to move away from VMs if required.

- Task Executor will be dual hosted, once as a Multi-Tenanted Linux Container, and once as a Single Tenanted Windows Container. The goal is to reduce the number of Tenants that need the latter provisioned but giving the option to move away from VMs if required.
    - Look at KEDA & Container Apps for Multi-Tenanted Task Executor (may not be viable due to Container App limitations)
- VGC and associated services will move to Linux Containers.
- Loyalty and associated services will move to Linux Containers.
- Task Executor will be hosted as a Single Tenanted Windows Container.
- Task Scheduler will be moved to be Single Tenanted Linux Container.
- Connect services moved to Linux Containers.
- Potentially use Windows Containers as a steppingstone and/or test mechanism.
- Service Framework Core Services will be containerised and released as such.

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Phase 3

- Some key IIS Services move to Linux Containers
- Sales Server Core will move to Linux Containers.
- Sales Server Customisations will move to a Windows Container.
- Core Transaction Services will move to Linux Containers.

Phase 4

- More elements are containerised. Ideally on Linux containers where possible.
- Service Framework Services move to Linux Containers.

Phase 5

- Continue last elements of containerisation

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Creation (and usage) of a new Extensibility Framework (7.5)

Problem

At present, there are a multitude of extensibility options for non-core workloads to be executed periodically. These workloads include extracts, imports, and reports and are usually operate as tasks, web services or as queue listeners.

Created by Vista's New Zealand Office, Vista's overseas offices, Business partners or in limited cases, Customers; these integrations have proliferated into the hundreds and are one of the primary ways that Vista's solution is able to operate to fulfill the business and compliance needs of customers in 100 countries.

As Vista moves into the cloud, these integrations start to come with several issues:

- Integrations need to be modified to operate in the cloud, with changes to date/time handling, connection strings, disk access, configuration, logging, and packaging. This creates challenges in development as well testing as the on-premises task scheduler is different from the cloud task scheduler.
- By creating a set of business logic in a task, that business logic is often not transportable to be triggered by other mechanisms (such as a web service call or queue event).
- Security and Licensing is often inconsistent and not well set up for customer and third-party consumption.
- Our queue listeners do not have side by side version support, meaning that a system upgrade for a single site has potential to break functionality for a full circuit.
- Database schema scripts often need to be run to allow data persistence in the DB. In a cloud operation world, this means that scripts need to run against all a customer's databases to run.
- Ill behaving integrations can negatively impact system performance (especially when being run for a lot of sites).
- On premise tasks run in a serial manner, thus tasks that operate both in the Cloud and on premise run at different cadences, creating architectural issues when designing and building solutions that need to run both on the Cloud and On Premise.

Ultimately, the current system, whist fit for purpose on premise, creates major barriers to productivity and operation in a cloud environment and is ultimately not fit for purpose in Vista Cloud. It is time that we create a new development paradigm that can allow our development teams to focus on developing business logic to solve customer problems, rather than being concerned about the plumbing and having to test everything twice.

Solution

We would like to have a robust FaaS framework in place that allows us to have a wide range of Software Developers (internal and external) create and deploy workloads that run in Vista Cloud in a simple, cheap and low risk way. We empower creating better software faster.

Phases

Phase 1

- No Progress

Phase 2

- Architectural Design to occur.

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Phase 3

- Base Classes Created and template created.
- One or more tasks converted to this framework.
- Legacy and Cloud Task Scheduler updated to enable Consumption of FaaS created components.

Phase 4

- Queue Tasks can use FaaS business logic
- API Enablement of FaaS business logic

Phase 5

- Further Refinement of Runtimes. Possible extension to allow for Azure Function Execution if financially viable.

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Observability Enhancements (7.6)

Problem

Modifying our software suite to ensure it is manageable in a SaaS/Cloud-deployed environment is critical to minimize the man-power resources required to maintain a high quality of service for customers. The existing software suite is Cloud-compatible and has had significant investment into observability, but still lacks some coverage required for effective operation.

When running on premise, the first indication of software fault is often a customer logging a ticket on the Vista services site.

When Vista's hosting the service, we need to be able to pro-actively be aware of issues and be able to resolve them quickly to reduce the chances of an SLA breach and keep a high level of trust in the service.

Solution

We would like to be able to identify and diagnose issues with any Vista piece of software quickly, such that appropriate action can be taken to resolve the issue. This includes Server based elements as well as desktop and web clients. This needs to cover error logs, telemetry, distributed tracing as well as audit trail information to give us the best chance of identifying and fixing the issue as fast as possible.

Phases

Phase 1

- Enhancements for visibility of Forms Launcher & App Launcher.
- Enhancements for visibility of Task Execution.
- Architecture and Cloud Ops/Engineering & Engineering work together on observability standards for Multi-Tenanted Core Services.

Phase 2

- Sales Server (Core and Customisation) and Transaction Services will have additional observability added as part of their move to containers.
- Task Scheduler and Core tasks will have refined observability to aid in diagnostic and performance analysis.
- Front end management websites will have fault detection and logging built in and will send diagnostics to a central location. Likely using a commercial offering in this space such as Raygun, AppD or Site 24x7.
- Enhancements for visibility of Loyalty.
- Enhancements for visibility of VGC.

Phase 3

- Customer facing Status page showing current issues and defects. Likely integrating with commercial offering in this space.
- All Services and Components moved to Containers and/or are Multi Tenanted have additional observability applied. Including Distributed Trace, Business and Performance Metrics as well as Log Cleanup.

Phase 4+

Ongoing work to improve and tweak observability across the suite.

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Configuration Enhancements (7.7)

Problem

Vista has traditionally had a multitude of ways to configure the operation of individual applications. These include:

- Maintaining configuration in a database.
- Maintaining configuration in an application manifest file (traditionally, web.config).
- Maintaining configuration in an ini file.
- Maintaining configuration in Vista's new Central Configuration system (DB backed, web service and queue-based value distribution). This deals with hierarchical configuration and configuration that differs on a per site basis.
- There is also a special configuration type that defines the connection strings to individual databases.

With Vista configuration settings, there is a fair amount of variation on the purpose of different configuration entries. Some of these configuration settings are for the purpose of software initialization (e.g. file paths), some others are for global execution behavior (e.g. Connection Pool sizing) and the majority are for software operation (e.g. URL of downstream services).

This creates a great deal of complexity when it comes to maintaining and modifying the system, creating issues through misconfiguration and difficulty resolving issues due to the disparate nature of these settings in a variety of places.

Solution

No configuration lives in files. Instead, configuration lives in:

- The product database (where the configuration is of complex structure and does not fit well into central config's tooling. Some configuration may be distributed from Central Config into the databases to stop large scale client changes (e.g. tblConfigure))
- Vista Central Config (For user/customer modifiable configuration)
- A Container Configuration system (for dev and ops teams)
- A feature flag management system (where the configuration is system level, but should be controlled wider than the dev and ops teams)

Ultimately, this is about cutting down on the effort required to set up and modify the system and reduce the chance that incidents will result.

Phases

Phase 1

- The control plane will be used for automated distribution of legacy configuration (see the Control Plane section)

Phase 2

- Core tasks will have all file-based configuration removed.
- Active work will occur to remove configuration elements that are found to need to be changed post initial deployment.
- VGC will be enhanced to remove all file-based configuration.
- Loyalty will be enhanced to remove all file-based configuration.

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Phase 3

- All file-based configuration is removed in Vista Cloud (for non-static elements).
- Key configuration is able to be migrated across environments.
- Sales Server Core and Customisation will have no file-based configuration.

Phase 4

- More configuration is enabled for environmental migration.

Phase 5

- All key configuration is transferable across environments.

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Move to favouring Extensibility over Customisability (7.8)

Problem

Over the last 20 years, Vista has continuously added features in its software to satisfy the ever-growing requirements of customers in 100+ countries.

Since Vista is a single source code base, the modus operandi thus far has been to add configuration flags or new entity fields to deal with the increasing level of complexity that comes with the broad level of geographic and business process coverage we have grown to encompass.

Whilst this has created a system that is incredibly flexible and adaptable, it has made Vista's software incredibly complicated, and thus has created several unintended consequences:

1) It is increasingly difficult and slow to add new features as the sheer number of integrated code paths and configuration options mean that development and test time is elongated to make sure that we have not created defects.
2) Test automation coverage is increasingly difficult to add as the number of unique combinations of configurations and data setups means that code flows can-not be covered to give a high degree of confidence that a change is safe.
3) Creation of functionality that is relevant to a single customer or region is serialised through a single team (or set of teams) in New Zealand.
4) This functionality came as part of the core software and therefore had no way to monetize the additional functionality or fund the maintenance work that comes from that added complexity.
5) We're limited in our ability to innovate, creating solutions that have to cover a wide range of customers and scenarios, without the ability to take risks with our capabilities to drive customer and patron engagement.

Ultimately, the flexibility and configurability of the software we have create has to be weighed up against the negative externalities caused.

Thus, we must begin to favour extensibility in our code execution paths that covers a specific business process, rather than configuration that tries to cover all of them. In addition, we need to revisit some areas of complexity that continue to slow down innovation and cause customer confusion and issues.

Solution

The majority of complex functionality that affects core transactional systems is delivered via software plug ins rather than more complexity of configuration. Ultimately, we want to reduce the risk of an upgrade and allow flexibility to bring in Marketplace participants to extend the Vista ecosystem.

To make this statement a bit more concrete , I'll give an example of how we have recently designed a solution that follows this paradigm, the benefits that we see and some potential retrofitting we can do to allow for future flows to be supported.

The Promotional Gift Card Savings work was created to address the needs of customers that wished to discount items such as gift cards (e.g. buy $50 in value for $40), but wanted to therefore reduce their film hire and tax liabilities as a result when redemption occurred, with the customer not seeing such complexity.

Several customers faced the same problem; however, each had slightly different requirements around the calculation of discounts and the items upon which they apply.

Traditionally, in Vista, this solution would have been completed with new payment type, item and system configuration flags. However, these flags would likely have been read within core POS and Sales Server code, integrated into existing code flows and therefore further complicated an already complicated area of Vista software.

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Instead of following this existing path, Vista will create a connector model, where the basket content can be passed into a discount connector, which will allow for in modification of tax values and the creation of compensating payment lines.

This has the following benefits:

- We are licensing this behaviour to customers through these connectors, making sure that we get compensation for any additional complexity that exists in our software.
- Only specific flows are supported, thus allowing the connector to be greatly simplified and testable in isolation.
- Future scenarios can be supported by Vista's regional offices as they are able to create this business logic without risk to core logic or other customers.
- Test automation is much simpler, as we only need to validate that models are correctly passed to a connector, and that the data is applied subsequently, rather than trying to validate all the combinations that a customer could come up with around discounts and limitations.

Phases

Phase 1

No progress

Phase 2

- POS uses stateless deals engine. This enables external plugins of both pricing and deals from sources such as CXM, allowing us to externalise this logic from core Head Office.

Phase 3

- Pricing now contained in an extensibility library.

Phase 4+

- New features are imagined through the lens of Extensibility.

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Create (and use) a Feature Flag Framework (7.9)

Problem

Over 20+ years, Vista has created incredibly complex software to service the needs of customers in over 100 countries. Our traditional release process has involved periodic shipping of feature releases to customers every four months, with an expectation that the customer would then have a (potentially months long) regression test process to validate that their unique configuration and integration sets were not impacted by changes we made.

It is a long, long road to getting adequate test coverage for 100 countries worth of customers and customisation and will likely never eventuate for all code paths.

However, with the above said, we would like to move to a model where we are able to deliver value to our customers more frequently, with changes made by our teams delivered potentially hours later to our customer base. Bluntly, these competing considerations are at odds.

Solution

In order to enable the increased frequency of deliveries to customers, we will lean heavily on the integration and mature use of a robust third party feature flag framework (such as LaunchDarkly) in order to enable the progressive rollout of new features, allowing for new code paths to be shipped in a disabled state and then progressively rolled out (and potentially rolled back if required).

Feature Flagging allows us to choose the best way to roll out features, whether that be through Data Centres, Tenant Rings, Release Rings or User Samples. We can then tie the rollback of these features to automated metrics sampling around increased errors (or not).

Phase 1

- No Progress

Phase 2

- Identification of Commercial Feature Flag Framework and POC. Include approvals workflows and automated roll out functionality.
- Introduction and Integration of External Feature Flag Framework.
    - Integrated into containerised and some key server-side workloads.

Phase 3

- Use Feature Flag rollout automation to allow for automatic rollout of features through the customer base.
- Audit and Approval Flows Introduced.
- Connect Telemetry to Feature Flag rollout process, allowing for automated rollback under certain alerting conditions and notification to key stakeholders.
- Add Feature Flag Framework into Front End Systems
- Integrated into strategic client devices. POS, Serve, Rove, Omnia Kiosk

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Phase 4

- Database Business Logic integration with Feature Flags.
- Ability to push feature flag values to target databases from Flag Management Control Plane.
- All Major new features are Feature Flagged.

Phase 5

- Further refinement of the Feature flag system from operational learnings.

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Licensing (7.10)

Problem

An ongoing challenge for Vista in an on-premises world is the difficulty to license individual products to customers. Apart from a very broad strokes licensing system that licenses an individual cinema to a specific number of screens, for many years Vista's licensing scheme had assumed that if you had access to a software package and the knowledge of how to install it; you were licensed to do so.

As Vista progresses in SaaS, that level of licensing is not appropriate. Rather, we need the ability to license both capabilities and access. Both for individual customers (and customer environments), as well as third party integrators who wish to access services that we host on behalf of our customers.

Solution

Vista software will create a sustainable licensing suite to enable both third party access as well as product capabilities in the Vista Software. Control of this mechanism can be delegated to the correct people in the organisation (with appropriate approval follows) and that billing systems are able to be integrated with a minimum amount of human intervention.

Phases

Phase 1

- No Progress

Phase 2

- Enhance licensing to smooth edges of release of Cloud only features to market. (No per customer enablement on premise)
- Licensing use cases extended to enable per customer enablement of Features.
    - Able to implement licensing for all new features targeted from this point onwards.
    - Able to license a cloud customer for a feature
    - Able to use CIM to distribute license credentials to on prem customers
- Begin retrofitting licensing into Vista Cloud, enabling/disabling functions based on appropriate customer licensing
    - Start with product enablement and move through smaller features.
    - Enable different licensing for different customer environments.
    - Enable export of reporting for analysis and billing.
- GAS is replaced with CIM for all purposes. (Pending verification)

Phase 3

- Self Service Licensing abilities are introduced
- Licensing of Vista Cloud Capabilities are completed.

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Phase 4+

- Further refinement of the Licensing framework from operational learnings.

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Software Usage and Deployment Telemetry (7.11)

Problem

One of the tremendous challenges that Vista faces in its product and engineering processes is the lack of good quality data with which to inform decision making.

We are constantly faced with questions that we cannot answer:

- What features are our customers using?
- How many sites use Kiosk and on average, how many do they have? Are they pickup or ticketing Kiosks? Do they sell concessions?
- How many customers have a configuration flag set to a specific value such as Cash Management Mode?
- How many people run this report?
- How often does someone interact with a specific entity?
- What plugins does a customer have for Fiscal, ERP, Payments, Printers etc…?

This ultimately means that we cannot make informed decisions on things like test automation focus, pricing, feature creation, product and platform obsolescence and many others.

At minimum, we spend a long time trying to understand the gap between on-premise and cloud, elongating the Cloud Journey for customers and slowing cloud adoption as we have to constantly ask the customer these questions and are beholden to their own information finding process. The accuracy and thoroughness of this process could be a barrier or accelerator for Vista Cloud adoption depending on how well we're able to do it.

Note that there is likely another project that needs to sit alongside this which is gathering specific user interaction with web interfaces so that user journeys and user experiences can be optimised. However, ultimately, the two projects have different data requirements and very likely, different tool set, so this is focused on a set of data that is most relevant to engineering that is not captured by alternative tooling.

Solution

We capture good telemetry information at an aggregated level to make business (e.g. Product Enhancements) and technology decisions (e.g. What test cases to cover).

Phases

Phase 1

No progress.

Phase 2

- Consider the question of on-premises telemetry to expedite the gaps-analysis and migration of customers to Vista Cloud. This could include elements related to
    - Plugin/task library usage.

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    - Report and Form Usage.
    - Analysis of differences from base schema to understand custom elements that customers have added to databases.
    - Version Usage
- Aggregated collection of key information from Vista Cloud configuration to drive product utilization and configuration. Key outcomes are:
    - Test Automation guidance.
    - Availability of good data for product teams to make feature decisions such as feature utilisation and configuration flag setup.

Phase 3+

- Further refinement of on-premises data collection for migration purposes.
- Further refinement of Cloud data collection for business decision making.

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Offline support (7.12)

Please note, that this is a product driven epic, and the phasing and execution details will be driven by the Product team. We have included this section for completeness.

Problem

In SaaS, there are a lot more potential reasons why Vista may go offline (compared to an On-Premises installation). These include:

1) Disconnection between site(s) and the Internet.
2) Disconnection between site(s) and the WAN of the Customer (if routed centrally).
3) Outages caused by our Hosting Provider (Azure).
4) Outages caused by Third Party Dependencies (such as Azure AD or Auth0).
5) Outages caused by Vista.

With a very regular upgrade cadence and a proliferation of sites on SaaS, we should expect that offline events are going to be an inevitability.

It is important for adoption that our customers have Trust in the platform, and at the heart of this trust is the ability to keep their most important workloads working when service is unavailable. This risk is the most identifiable one moving to the cloud for our customers and is often discussed in any conversation with a potential prospect.

Solution

Customers will feel comfortable bringing their workloads to the cloud, with the ability to apply operational processes backed by robust software to allow in cinema trading.

Some customers may still want to retain some on-premises function for comfort, and thus we have left ourselves open to creating a managed appliance offering (using Azure Arc) for these types of customers in later phases if needed to accelerate platform adoption with customers who have a large aversion due to missing offline capability.

Phases

Phase 1

- We have received feedback from some customers that POS tries “too hard” to stay online, particularly in cases of network slowdown vs. outright network outages. A preferable approach here would be to allow customers to manually go into offline mode in cases of performance issues. The work includes:
- POS already has a range of offline support, however customers are often not aware of what POS can actually do offline when enabled. Whatever we end developing within this epic, we should deliver updated documentation that clearly outlines to users what they can and can’t do while offline.
- We have also received feedback that has suggested it isn’t immediately clear that POS is doing something when it has a poor connection with the server. While it may look like POS is stuck or frozen, POS is waiting on a response from the server before continuing. Because of this people can press buttons more times, and slow down POS even further or cause it to stop responding.
- Checks are not viewable while offline. This means that if a customer wishes to pay for a check, they are stuck. Their only option is to re-ring in the transaction – viewable at a channel like Serve – as an offline concession sale.

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However, to do this they need to re-request the customer's card and run it out to POS. These checks can also not be printed. This work will enable a read only view of cheques when offline.

- POS can be configured to enable the collection of bookings while in offline mode. This functionality is beneficial as it allows orders that have been taken in advance of POS starting to be collected to be found and collected – when not enabled, tickets and concessions made online cannot be collected and must be validated operationally. To allow this functionality, whenever POS starts up it loads all bookings that have been made for that day and stores them locally. However, this functionality is rarely used due to the increased time it takes for POS to start up when enabled – commonly 45 seconds or more. Improving performance of this process would make customers more likely to use it.

Phase 2

- While offline, POS proceeds into unallocated seating mode. The offline seating service – developed a couple of years ago – was built to allow POS to continue selling allocated seating while offline, with screens assigned to individual POS terminals to prevent overselling. We will need to enhance this solution so that POS Terminals communicate with one another and share seat maps and counts.
- The Kitchen Printing Service is a Windows service that reads values from the Kitchen tables and pushes these through to network printers in real time. As POS is not directly connected to these printers, when POS is offline – and not writing to the kitchen tables – these printers will silently fail. When in offline mode, POS should print directly to a designated network printer – assuming that POS can connect to the LAN.
- Many F&B customers run their POS terminals in checks-only mode. When offline, this renders them useless. Once in offline mode, POS should be configurable to automatically enable standard concession sales.
- While offline, the majority of buttons at POS are disabled. Several of these are critical to F&B operations, and have little reason to be disabled while offline. The most relevant of these are Set Delivery, which allows a user to specify delivery information (time of orders, order comments, order priority etc.).
- POS' Tips screen only displays tips for the current business day. While you can add a tip via a payment's Authorization Code, POS only searches for tips across the current business day. Assuming that settlement has not been run, POS should allow for tips to be altered via authorization code for previous business days.

Phase 3

- Offline support addition for common Fiscal/Ticket/Receipt Numbering situations as it relates to major regions.
- Once checks are viewable offline, we should also be able to pay for them too. While we cannot reach the database while in offline mode, if we can view a recent version of the check we should be able to "replicate" this order into a normal transaction that can then be paid off.
- While POS does not currently support opening checks while in offline mode, Serve does – sort of. It allows you to open any checks you previously had open on that device, as Serve doesn't have an "offline" mode like POS does, and caches checks and similar data to improve load times.
- Consideration of offline support for Kiosk Booking Pickup, Navigation and simpler use cases

Phase 4

- Consideration of offline support for Kiosk Extensions.

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Phase 5

- Further refinement of Offline support in Client Devices and/or managed appliances.
- Consideration of Managed Appliance release based on Azure Arc if required for acceleration/fulfilment of adoption curve.

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Backwards Compatible Schema (7.13)

Problem

At present, software is upgraded in one serialised process, with the upgrade of Client and Server software linked to the upgrade of DB schema.

This means that if critical issues are found that we cannot fix in a short period are found, we do not have the ability to roll back to a previous "known good" version of the software.

Solution

We can make database upgrades to customers environment in a safe manner before a customer's application software is upgraded to a new version. This can be done far in advance of associated software change.

On discovery of an issue in an upgrade or feature enablement, the schema will still be able to operate without fault for the reverted version(s).

Phases

Phase 1

- Documentation is created for best practices for creating backwards compatible schema changes. This is distributed and training done with all engineering teams.
- Database deployment from nuGet packages. (see 7.14)
- Separate installation of Stored Procedures, Triggers and Functions from core schema, allowing for rollback of business logic.
- Test Automation for PIT for breaking change analysis.

Phase 2

- Test Automation for all teams of Breaking Changes in Schema. Likely for testers as well as builds.

Phase 3

- Refinement of Schema Validation tooling as required from operational experience.

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Modernising our Database Deployment Software (7.14)

Problem

At present, our DB deployment, whilst progressed from that on-premise, is not up to the task of upgrading potentially thousands of sites in a controlled way across the world.

Solution

We have a fit for purpose, robust way to roll out database changes to hundreds of databases around the world in a low touch manageable manner.

Phases

Phase 1

- Database updates coming from nuGet Package representations of package constituent parts. Allows for restructuring of databases into separated packages.

Phase 2

- Work to extend our database deployment technology after learning of deficiencies in phase 1 experience. Especially at higher scale.

Phase 3+

- Further refinement after operational experience.

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Restructuring our DB Script Source and Release Process (7.15)

Problem

Vistas traditional mechanism for storing and modifying database scripts was in the TFS repository, creating a singular space where multiple teams are delivering all scripts for all features.

In addition, at release, these scripts are all dumped into a single place, losing any links between the product areas and the associated schema changes, meaning a functional upgrade needs a system wide schema update also.

Solution

We split our Database scripts into segregated bounded contexts with robust processes on top to allow for our teams to own and manage their relevant schema elements. For example, putting in place a PR and review process for Schema areas.

This would allow schema upgrades for specific areas, allowing for us to decouple release cadences for various product areas. When combined with a more robust upgrade process and backwards compatible schema and many of the application changes in this document, it allows us to push features with confidence to customers much more regularly.

Phases

Phase 1

- No progress. 7.14 gives a clear path to restructuring of databases into separated packages.

Phase 2

- DB Scripts start to be moved to Azure DevOps and released as NuGet packages. This includes:
  - Structure for team/product permissions for change management.
  - Cloud publishing of packages for inclusion in automation tooling.
  - Re-publish/constitution of scripts into on-prem systems.

Phase 3+

- Further refinement after operational experience.

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Moving to Unicode only databases (7.16)

Problem

Currently Vista products support running on both ASCII (non-Unicode) and Unicode character sets. The choice is made when the database is created and is expected to be consistent across the customer's circuit.

The current migration process duplicates this decision into our cloud environment, and thus the dual target creates several performance and support issues as our software, teams and processes need to be cognizant of two different storage worlds, even in Vista Cloud.

Solution

We migrate customers so that all customers have the same schema applied to their databases, reducing support costs and allowing us to tailor performance testing to a known target.

Phases

Phase 1

- No progress.

Phase 2

- No progress.

Phase 3

- Tooling begins to be created to migrate DBs to Unicode DB in situ for Vista Cloud.

Phase 4

- All databases converted to Unicode as practicable.

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Central Identity Managment (7.17)

Problem

In the past, Vista's Identity Management could be best be described as "ad hoc", where User repositories were created and spread across a great number of individual products with no robust SSO support. Thus, a single user may have half a dozen or more disparate logins to keep track of throughout the product suite.

Vista Cloud's Central Identity Management Suite is a massive improvement in this situation, creating a robust, secure mechanism to secure user access to our products with a greatly improved user experience and a single login that enables access to all.

However, this is only a first version, and there are a number of products yet to be integrated and key functions yet to be implemented.

Solution

All product interactions in Vista (outside Operator Devices) are through a single identity. We have a clear and coherent way to manage users and their permissions throughout the Vista Software Stack.

Phases

Phase 1

- Most products are CIM enabled, including those which require workstation binding.
- CIM Services set up for High Availability.
- CIM Services are set up in 2 or more regions.
- Enable mixed user repositories per Tenant (Federated and Vista Managed)
- Multi-Language Support.
- Provision User Access Privileges before staff onboarding.
- Modify configuration to support increasing Tenant Count

Phase 2

- Improve the user experience of managing users from a single location. Including the ability to define User Groups rights within the landing page.
- Add additional user level authorisation rights into the User Management pane of CIM (Legacy User Maintenance for various apps).

Phase 3

- Further refinement after operational experience.
- Manager Delegation of User Rights Access Control for their Reports

Phase 4

- Further refinement and improvement after operational experience.

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Increasing Test Automation Coverage (7.18)

Problem

One of the ultimate goals (and differentiators) of Enterprise SaaS is the ability to quickly roll out new features to our customers and have them using them soon after our engineering teams have completed them.

With multiple decades of feature development, a great deal of it without test automation coverage, we are currently in a place where increasing velocity of feature releases in production can come with significant risk.

Whilst elsewhere in this document we have described a multitude of compensation mechanisms to make sure that we are able to quickly identify issues and roll back to a previous version (or forward to a patched version), all these mechanisms are predicated on a visible failure occurring.

Ideally, we'd capture these failures before a deployment step and the issue had impacted a customer.

Solution

We greatly improve our test automation coverage in critical areas such that issues that most impact customers are found before release via automated test coverage. This includes performance and security regressions.

Phases

Phase 1

- Further maturity of Performance Test Engineering Suite. Especially as it relates to the Sales Pipeline and background processes/tasks that can impact the performance of the Sales Pipeline.
    - Planned test coverage is therefore designed to include
        - All Connect and OCAPI load
        - All Loyalty load
        - All Cinema load
        - All VGC (lower priority, as not normally critical transactions)
- Automation of creation of Developer, Test and PIT environments using methodologies close to that of Production.
- Creation of Future Test Automation Phase plan in conjunction with Test Automation Leadership.

Phase 2

- Execution of Test Automation Plan as designed by new Test Automation Leadership
    - Focus on areas of high risk and/or areas where we wish to increase release velocity.
- Execution of Test Automation Plan as designed by Performance & Test Automaton Leadership.
- Design of and Execution of Security Test Automation Plan designed by Security Architect and Test Automation Leadership.

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Phase 3+

- Increase of Functional, Performance and Security coverage as designed in Phase 1 & 2

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Control Plane (7.19)

Problem

The way that Vista currently deploys customer specific customizations in Vista Cloud (such as Payment Connectors and Fiscal Drivers), is manual, difficult, and often results in mistakes that can cause production outages. Bluntly, the current system will not scale with our customer adoption if it requires manual intervention on a per-tenant and/or per upgrade basis.

Solution

By prioritizing command and control elements and allowing it to operate over both Container and Virtual Machine deployments we can reduce support costs and outages, as well as have our teams using a consistent method to upgrade software (now and into the future).

We'd like all Tenant Specific changes to application software and databases are managed by a robust Control Plane. It should allow for role and environment specific rights, full auditability, the ability to "promote" this across environments, and the ability to roll back an environment if an issue is discovered.

Phases

Phase 1

- Productise control plane and database deployment "Dane's tool".
  - Basic customisation storage and deployment. Stored in Git with storage model close to target layout.
  - Create VM images containing OS, Vista software, and third-party software. These VMs correlate to servers in the existing Vista architecture: Head Office, Loyalty, VGC and Cinema. The images can be reused for all tenants.
  - Use the core software VM images to create VM images customised per tenant, with all their configuration applied and customisations installed (as per the Git Repo).

Phase 2

- Core Component Repository Created with links from Tenant Repositories.
  - Allow for common extensions to be staged once.
  - Allows for replacement of config values to enable separation of base configuration from environmental configuration.
- App Library customisation supported more robustly (as per above link/promotion abilities)
- Environmental export is enabled (for testers/QA).
- Audit and Approvals flows in place.
- Database Scripts can be executed.

Phase 3

- Environmental promotion is enabled.

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- Integration with Marketplace/external control elements to enable self service component installation/upgrade by a customer.
- Customer visibility of installed components and history.

Phase 4

- Further refinement after operational experience.

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Shared Queue and Cache Resource capability (7.20)

Problem

One significant cost for any VMS deployment is the cost of dedicated queue and cache resources. This cost is exacerbated with customers not being able to share these resources between their Production, UAT and Test systems.

This Architectural Limitation will get very, very expensive over time if not resolved.

Solution

All Queue and Cache resources are made to be shareable across Tenants.

Phases

Phase 1

- Redis and RabbitMQ can be shared.

Phase 2

- Queue Processors can be shared, including Task Scheduler & Executor.

Phase 3

- Further refinement as required.

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Database Scale Support (7.21)

Problem

Vista's Server software has come from a world where the Database is co-located with the software, often with the database on the same machine as the services that use it.

With the move to the Cloud, the Database Servers we use are usually PaaS based. That means that they can go offline while the services themselves are operational. However, more importantly, on these cloud platforms, the scaling of a database/pool/server results in a connection break for open connections.

Our software does not handle this gracefully, with failures and potential data integrity issues as a result. This results in us often holding a larger balance of SQL Server capability than required to minimize the frequency (and thus impact) of scale events. This therefore maps directly to cost.

This issue is exacerbated with SQL Azure Hyperscale Elastic DB Pools (currently in private preview), which would allow us to match compute capacity otherwise programmatically to current need (and thus further incentivize aggressive scaling).

Solution

We have modified our software to enable us to scale elastically often, and with confidence.

Phase 1

- No progress.

Phase 2

- Some work in common Vista Libraries to adapt to SQL Azure specific availability events related to scale. Focus on Services the Transactional Service Hot path.

Phase 3

- All Transactional/Critical workloads changed to adapt to a scale event.

Phase 4+

- Additional workloads changed to adapt to a scale event if any manual change required.

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VPN Removal and Network Improvements (7.22)

Problem

At present, the vast majority of Vista's products and services have not been hardened in a way that allows for usage of the system by customers without VPNs. This is a barrier to adoption and takes away from a true SaaS feel to the product.

In addition, we need to create a mechanism for traffic to be secure within the data centre, flowing between VMs and Containers, in both single Tenanted and Multi-Tenanted configuration.

Solution

We modify our software such that a VPN is not required for any day to day operations of our software. A customer may choose to pay an additional fee for a VPN, and we *may* have a VPN for those customers that still retain some Vista Server Components on premise (such as one or more Cinemas).

We will have a robust mechanism to create cross system and cross service communication that allows for workloads to be moved incrementally, in a secure and performant way.

Phases

Phase 1

- POC/Groundwork on Service Mesh ready for implementation in Phase 2.
- Phase 1 of Network Ingress changes (potentially on App Gateways or Similar)

Phase 2

- Remove VPN requirements for Loyalty & VGC
- Removal of API-Router from all calls in Vista Cloud, to be replaced with Service Mesh.

Phase 3

- No SQL Execution Service for POS.
- Removal of DBEngine Web Service from all workloads.
- The requirement to use a VPN to access Vista software is removed for all Cinema and Cinema Users.

  Pen-Test undertaken on Cinema VPN removed workloads.

Phase 4

- The requirement to use a VPN is fully removed when sites are cloud only.

  Pen-Test undertaken and changes applied.

Phase 5

- Pen-Test undertaken and changes applied.
- The requirement to use a VPN is removed for customers that still have sites on Premise.
- Pen-Test undertaken and changes applied.

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Bridging SaaS Components on Premise (7.23)

Problem

As Vista's SaaS solution gains market share, it is likely that a large percentage of Cinemas will move from On Premise to the cloud.

Depending on a dozen or more factors, many out of Vista's control, it may be possible that we have a substantial set of customer sites that remain on premise for some time to come

However, it's likely that a large percentage of those will have Central modules running in SaaS and/or wish for Vista to manage their on-premises systems, or at least have their cinema components update such that they are still compatible with the Cloud Modules.

Thus, depending on how things evolve, may need to look at enhancing the way we deploy on premises cinema systems to adapt to this commercial reality.

Solution

Vista's on-premises deployments have been adapted to enable commercial opportunities related to Vista Cloud, Vista Digital and Vista Marketplace.

Phases

Phase 1

No progress.

Phase 2

- Some work on deployment of Service Framework Services in a simpler manner on Premise to enable Vista Digital and Marketplace updates.
- Automatic deployment of database scripts that go alongside these Service Framework Services.

Phase 3

- Mature distribution of components related to Vista Cloud, Vista Digital, Vista Horizon & Marketplace.

Phase 4

- Further refinement after operational experience.

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Desktop Software Transition (7.24)

Please note that this section is included for completeness and is managed primarily as a product initiative.

Problem

At present, most data maintenance operations at Cinema are still done primarily within the Back-Office application. This means that users are often working in two products at the same time, requiring knowledge of our own migration progress to understand where to complete tasks. The same will be true in future with our other Cabinets applications such as Head Office and Voucher Manager

With Vista Cloud, we need to move all of these workloads to a web interface.

Solution

All Operational Maintenance and Reporting is done via a Web Based User Interface. During the preceding period the Desktop Launcher is used to facilitate the incremental movement to this end state.

Phases

Phase 1

- Some progression of Back Office to Cinema Manager forms as needed. Some “big” common forms transformed along with a few smaller ones.
- Work to improve Desktop Launcher. Including Data Access Service Connection sharing.

Phase 2

- Cinema Manager Master Data forms multi-targeted to Head Office Support.
- Continued investment into Master Data Forms, through a mix of Complex and more simple data forms. Focus on forms with high latency impact on customers. Geographic distance from data centres is likely to be a barrier for some customers and the most impactful area is likely commonly used Maintenance forms.

Phase 3+

- Continued investment into Master Data Forms, through a mix of Complex and more simple data forms.

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Deployment and Upgrade Automation (7.25)

Problem

With the move to containers, the current installation process is no longer going to be work, and a new process needs to be put in place. Much of the groundwork for this has been laid with Vista's Connect SaaS product, however, Cinema is unique in that we have the same system servicing multiple sites at once, potentially with multiple simultaneous "live" versions.

Solution

A robust mechanism will be created that allows an automated upgrade of Application Software and corresponding Routing Rules to facilitate a staged upgrade across our customer base.

Phases

Phase 1

- All Virtual Machines are stamped for base images. No Tenant specific images are supported with all customized elements to come from the Control Plane above.
- A windows Process is created to enable regular deployment of updates from source repositories during execution.
- All current Cinema Workloads still in VMs, but side by side support is removed.
- Connect Deployment is optimized to simplify deployment for new Tenants.
- Connect is similarly enabled for runtime update of extension elements if they change.
- Opportunity now exists for a "all in one" VM to allow sales to small circuits if required.

Phase 2

- Deployment streamlined for all Containerised workloads.
- Small refinements for VM based workloads based on operational need.

Phase 3

- Deployment streamlined for all Containerised workloads.
- Integration with Feature Flag Framework to enable rollback/forward operations by wider staff.
- Small refinements for VM based workloads based on operational need.

Phase 4

- Deployment streamlined for all Containerised workloads.

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Resource Utilization Improvements (7.26)

Problem

There is a direct relationship between the efficiency of our software to the costs that we incur in a SaaS world, especially within the database and application servers. By making targeted changes to our software that are now enabled by cloud native software, we would be able to drop the costs to host and deliver this software.

We have several areas of our software where we use far more resource than we should to accomplish common tasks.

Solution

The goal is to reduce our infrastructural resource utilization through targeted fixes and architectural changes outside those outlined in this document. We'd like to see a CTS approaching 25% of Revenue.

In many instances, this would also result in better customer experience as common tasks and processes would be completed quicker, even if there is a major blockbuster on sale.

Phases

Phase 1

- Minor fixes in place to reduce DB and app load.

Phase 2

- Implementation of some "low hanging fruit" decreases in system utilization through changes to key code and SQL.
- Non-Code changes rolled out such as Forced SQL Parameterisation.
- Purging set up for cleansing for performance purposes.

Phase 3

- Head Office Transition to SQL Azure
- Implementation of larger changes to reduce CTS and resource utilisation. This includes:
  - Targeted focus on Sales Server Temporary Tables.
  - Changes to place POS static data downloads behind a service.
  - Cache of common static access workloads (such as Cinema Manager Homepage)

Phase 4+

- Ongoing Investment Profile to reduce CTS costs, especially as we start to understand the barriers to CTS reduction of a much wider customer base.

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Additional Regional Extensibility Coverage (7.27)

Problem

As part of our original move to support Cloud, we made several changes to a lot of products to allow them to operate in the cloud. These changes were primarily focused in working in an environment where multiple cinemas (that could represent multiple time zones) are processed in the same software simultaneously. However, there are other areas where assumptions around connectivity or access rights are assumed and will no longer be true.

Solution

Vista's adoption curve must be enabled by our ongoing enablement of additional country and market requirements.

Phases

Phase 1

- Regional Coverage for customers added in this phase.

Phase 2

- Regional Coverage for customers added in this phase.
- Some common Integration components made cloud enabled.

Phase 3

- More work to unlock strategic regions/customers.
- Work related to the offline support added in Fiscal/Receipt/Ticket Numbering in 7.12.

Phase 4

- More work to unlock strategic regions/customers.
- Regional Coverage for customers added in this phase.

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Changing the way we deliver Client Software (7.28)

Problem

With the move to the Cloud, a large amount of our software will no longer be hosted on machines located on premise. However, there will still be Windows Client applications that persist into the future, including Point of Sale, Kiosk (even if the UI will be based on OCC) and Cabinets Forms.

Thus, we need to be mindful of the implications of increasing our release cadence to elements of software that are still deployed on premise. Our software has deficiencies in this area and these issues include:

- Network utilization by system updates at Vista and our customers can cause great cost and outages for us & our customers.
- The lack of roll back ability if a release is taken that causes issues.
- The ability to force an upgrade.

Solution

For client software that remains on premise (such as POS and Kiosk), we must be able to deploy new versions regularly with low risk of the change affecting Vista or Customer networks and enable the roll back of this software if/when an issue is encountered.

Phases

Phase 1

- Reduce impact of Software updates on Customer WAN environments, especially for POS updates.
- Facilitate rollback of POS for critical scenarios.

Phase 2

- Delivery of software from Static CDN locations to reduce impact on Vista Back end Resources.
- Further download sizes through Source Code/Package/Build restructure.
- Ability for client software to detect change required to enable roll-forward and roll back without human intervention.

Phase 3

- Peer Distribution of large files, to enable a single POS to download a version at a large site and then distribute locally. Use mechanism introduced for offline allocation of seats in previous year.

Phase 4

- Further refinement after operational experience.

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Vista Marketplace (7.29)

Problem

Vista's offering on premise has been wildly successful for a wide range of reasons. However, two of those are

- The wide range of products and functions that Vista has created for the Cinema Exhibition Market.
- The extensibility of the product allowing for customers, their development partners and the wider software exhibition ecosystem to extend the product for their own purposes.

In the future Vista is likely to continue creating a wide range of products and features for its customers upon the SaaS platform. However, it will also allow Vista Group Companies and trusted Third Parties into the equation, allowing these to be treated like first class citizens in the Vista Eco-System.

Solution

Third Party Integration, and much of Vista's own Customer Customisation will utilise API's hosted and managed on Vista's API Marketplace. The Marketplace will allow for secure access, a wide range of functional use cases and the ability to control and charge for third party access to Vista Cloud Customers, Veezi Customers and On-Premises Customers.

Phases

Phase 1

- Document and Categorise APIs
- MVP Created
    - Provide a strong, consistent authentication that allows all API consumers to be trusted throughout the API chain and allow Vista to approve and manage the consumers.
    - Expose the first batch of interfaces to deliver the business capacities and business data to our customers.
    - Have a single point of documentation of Vista Cloud platform for consumers and integrators to find all the relevant information.
- Horizon Data Sources Supported
    - Allow for the replacement of direct SQL integrations for data, enabling many more customers to be onboarded and third parties integrated.

Phase 2

- Support API accessibility with on-premises software and extend API cases
- Allow deprecation of v1 and v2 end points in Connect
- Begin Migration of Sales Server workloads to API platform.
- Veezi now exports to Horizon, enabling Marketplace participants to also target Veezi Sites.

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Phase 3

- Complete migration of Sales Server capabilities to API platform.
- Additional API's created and distributed.

Phase 4

- Further API's created and distributed.
- Enable distribution of Licensed Extensions through the Marketplace (such as Connectors/Tasks et al).

Phase 5

- Further refinement after operational experience.

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Vista Cloud Migration Software (7.30)

Problem

The migration to a SaaS environment will be a massive undertaking and will have to be done one piece at a time, likely over a period of weeks, months, or years.

The amount of complexity involved is often compared to changing tires while a car is moving. i.e. Business operations do not stop just because you want to move some of your workloads.

A massive amount of analysis, project planning, custom development, communication strategy creation, migration execution (often changing dozens of other systems at the same time) and post migration monitoring will need to be done for a project of any reasonable size to succeed.

It could be reasonably argued that each migration effort is as large/if not larger than a new on premise deployment as we are coming from a system state that may often have decades of data, configuration and customisation applied to it, creating a very bespoke problem for each and every customer and system encountered.

Solution

Vista has a robust set of tools in place to facilitate the migration of on-premises resources to the Cloud in a low cost and low risk way.

Phases

Phase 1

- Enable Import of Cinema User Data into CIM.
- Customer Migration for Connect.

Phase 2

- V1 Tooling that smooths commonly encountered barriers to Cloud Onboarding. Suspect this includes the automation of:
    - URL changes.
    - Config Extraction and push to relevant Vista Cloud location.
    - Data Purge (as required).

Phase 3+

- Further work to accelerate and smooth adoption from previous operational experience and customer adoption plans.

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Source Code, Package and Build Restructure (7.31)

Problem

One major bottleneck to productivity in the development team (and therefore productivity) is the time it takes from a developer making a code change to the point where that change can be deployed and tested.

In the future, one key element of our success with Enterprise SaaS is the ability to quickly deploy changes to an environment, whether that be because we want to deliver features, or that an issues has been found that we want to patch quickly.

Solution

Vista can Build, Package and Release software to production environments within two hours to facilitate fast fixes, as well as continuous delivery of value to our customers.

Phases

Phase 1

- Build migration and enhancements
- Work begins on setup of projects/templates for future storage in Azure DevOps.

Phase 2

- Accelerate move to Azure DevOps and NuGet.
    - Focus on items that are required by current Multi-Tenancy and Containerisation epics.

Phase 3

- Accelerate move to Azure DevOps and NuGet.
    - Focus on items that are required by current Multi-Tenancy and Containerisation epics.

Phase 4

- All source code is located in appropriately scoped Azure DevOps projects. (Perhaps GitHub if identified critical gaps are closed)
- On-Prem software packages can still be created, but we are optimised for C/D to Vista Cloud.

Phase 5

- Small changes as required as we learn from previous experience.

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Increasing Auditability (7.32)

Problem

Security and Trust are cornerstones of any SaaS service. In addition to the many other measures that are required to secure a SaaS service (some of which are included in this document), our customers need clear record keeping on the access and modification of data in their systems.

For example, as part of the requirements of GDPR, there are several use cases that are covered by having more robust logging in place. These include:

- Article 25 (Data Protection by Design and Default) – A data protection implementation must have the ability to prove manageability and traceability of actions taken against data in the processing environment. These audit logs must track the actions of end users as well as privileged or administrative users.
- Article 32 (Security of Processing) –To prove that security controls exist within the data processing environment, audit logs can provide documentation that those controls are in place and functioning properly.
- Article 33 (Data Breach Notification) – Organisations need to be able to capture security events in the form of audit logs to be able to confirm if a breach has taken place, and if so, how to measure the impact of that breach and determine what needs to be reported to the DPA and, ultimately, the affected data subjects.
- Article 17 (Right to Erasure) – When a data subject requests that their information be purged, the same type of visibility that detects a data breach can also be used to validate that this has been done and communicate this to the requestor.

Thus, Vista must expend some effort in creating a more comprehensive audit system that tracks, in much more detail how users interact with data in the system, especially as it relates to personal data, and ideally for other sensitive data objects.

Solution

Vista has modified its software so it can track the access of, and modification of key data, allowing us to fulfill the legal and compliance requirements of our and our customer's business.

Phases

Phase 1

- Connect adding specific headers to enhance Loyalty Audit.

Phase 2

- Audit Records are added as required for SOC Certification

Phase 3

- All systems have audit records generated when data changes back to individual users and/or system processes.
- New reporting will give visibility and searchability for these audit logs. This may not be in Horizon due to storage cost considerations.

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Phase 4

- Sensitive data access is logging into audit logging similarly to the data change events and exposed through the same reporting plane.

Phase 5

- Additional operations/access logged as required.

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Modernizing our Data Estate (7.33-7.36)

Problem

There are three key areas of concern

1) The vista report processing pipeline is aged. It has been designed with various separate goals in mind, and is scattered over multiple services, with various amounts of overlap between services. Whilst it does its job, the disparate engines doing various things makes a central management function difficult to achieve.
2) In addition, for most paginated reports, Crystal Reports is the rendering engine. Crystal reports itself is a powerful reporting engine owned by SAP. However, the licensing has become quite onerous, preventing Vista from upgrading to later versions, as well as the fact that Crystal uses a closed report format and has not progressed significantly since 2008. Currently, Vista uses Crystal Reports 2008 (Assembly version 12) which is not cloud optimized and limits our reporting capability.
3) The Vista data dictionary is a document which is available in various versions in various places. It is not easily accessible, and incomplete. It is not interactive, and does not allow data lineage to be followed.

Solution

Our Data estate is modernised to allow for the extraction of data in a modern, documented and audited way.

Crystal Reports has been phased out and replaced with an alternative that can run over Horizon and service paginated data reporting requirements for our customers in a much more flexible and fully featured way, including the ability to use these data sources for programmatic access.

Phases

Phase 1

- Some progress as it relates to data access through the Marketplace for Horizon origin sourced data. This becomes the accepted mechanism to extract/retrieve data from Vista & Vista Cloud.

Phase 2

- Begin process of migrating common paginated reporting to Horizon and off Vista Database.
    - Look at relative costs of Stored Proc changes vs Compat Layer creation.
    - Compatibility layers may be useful for other workloads, so consider this.
- Begin Data Dictionary.

Phase 3

- Continue process of migrating paginated reporting
- May need to add additional functionality to support scheduling/emailing.
- Continue creation of the Data Dictionary.

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Phase 4

- Continue process of migrating paginated reporting and data access.

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Break up our Large Front-End Applications (7.37)

Problem

The size of some of our applications such as Cinema Manager have grown so large to the point that they have become cumbersome to develop against and slow for users to consume, costing developer and tester time and good will from our customers that get frustrated with slow applications.

In addition, we would currently struggle with integrating extensions to functionality that are required to our software for regional and compliance requirements. Meaning that our remote offices and business partners have no clear path to integrating such software that used to be developed against the Cabinets Framework

Solution

We can ship updates to our large front-end applications regularly, allowing for bounded contexts to be updated without impact on other parts of the application. Remote Offices can ship parts of these front-end applications to meet local requirements that need a User Interface.

Phases

Phase 1

- Most common libraries and applications now in MonoRepo, reducing development times and decreasing time associated with release of new Angular updates.

Phase 2

- No progress.

Phase 3

- Start breaking up major applications into scoped areas.
    - Update builds/release to combine.

Phase 4

- Continue breaking up major applications into scoped areas.

Phase 5

- Further refinement after operational experience

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Changes to Head Office Downloads (7.40)

Problem

This architectural history means that we have accumulated a number of constraints on just how much we can change this system. But as sites progressively move to the cloud, customers (rightly) have an expectation that the concept of a "download" is an outdated one, where we are asking customers to download data from the cloud to the cloud.

That said, the accumulated constraints do place some elements of caution around just how far we are able to re-architect the system as there are a few considerations to take into account when determining the best course of action.

Solution

The Head Office Download Process is streamlined to largely remove the performance and operational issues that are created by the current download.

Phases

Phase 1

- No progress apart from those driven from customer issues encountered.

Phase 2

- Changes to resolve common download issues, including the size of downloads as they relate to Media.

Phase 3

- Further changes to resolve common download issues.

Phase 4

- Further changes to resolve common download issues.

Phase 5

- Major changes to HO Downloads to enable real-time updates and/or client changes to move master data sources.

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Changes to Head Office Uploads (7.41)

Problem

Uploads, like Downloads have evolved over the last twenty years to become the complex beast that it is. Aimed at periodically uploading relevant sales and reference data from site to Head Office, with data stored in summary formats, it suffers from several issues, including data mapping, timeliness of data as well as the granularity of data for reporting purposes.

A few years ago, Vista essentially re-created the Head Office Upload process from scratch, moving to a world where data is continuously streamed from multiple sources (including Head Office for reference data) to a central database and stored in full fidelity. Whilst a fantastic leap forward in technology, capability, and accuracy, it does leave a situation where two pipelines are reading the same tables and sending them to different sources, in different ways.

Solution

Head Office Uploads have been changed to enable a single, consistent data extraction process from Cinema to a Central Location. Horizon will then be used to distribute required data to the systems that require this updated Cinema data.

Phases

Phase 1

- No progress.

Phase 2

- Care and Maintenance changes only.

Phase 3

- Begin migration of HO uploads to use Horizon as a source for elements that will be retained in Head Office (as applications/reporting will require this information).

Phase 4+

- Continue migration of HO uploads to use Horizon as a source for elements that will be retained in Head Office (as applications/reporting will require this information).

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Vista Digital Platform and Channels

This document does not explore greatly the Digital Channels or Vista Digital Platform suite in any great depth, as this is better explored in a sperate roadmap.

However, for some level of completeness, I have included several identified gaps/work required at a Platform level for both VDP and the Sales Channels below.

Phases – VDP

Phase 1

- Connect/OCAPI: Running on Containers Stability.
- Connect/OCAPI: Professional API docs for integrators
- Connect/Web Loader: Load whether an MXF film trailer exists for a film and return this on the OCAPI film endpoints
- General: Improve/simplify deployment and configuration for cloud
- Lumos CMS: CIM Integration to allow customers to log into CMS

Phase 2

- Connect Manager is now in the Cloud and enables self-service elements for customers for the Digital Platform.
- Support for Connect Data Stream V2 (and removal of Web Loader apart from periodic refresh loads).
- Replace GAS with CIM
- External Authentication for Loyalty/2FA (external authentication replaces Loyalty & Connect for users)
- Security Audit
- Connect/OCAPI: Smarter cache invalidation (e.g.: set long lived proxy cache lifetimes and invalidate via CloudFlare API during web loader updates)
- Cache Control, Browser and CDN… Change to short cache, but as long as possible on CDN, and force invalidation when data change occurs.
- WPM/Connect CDN: Decouple from Connect installations/deployments
- Connect: Consider using SendGrid's templating or Movio functionality to author emails.

Phases – Sales Channels “Platform” Improvements

Phase 1

- Lumos/OCC: Nx Monorepo to reduce build/test times
- Lumos Web: Server-Side Rendering - preload layout HTML and GAS token & CMS API calls
- Lumos Web: Server-Side Rendering - image optimisation [need to also consider MXF CDN here]
- LT: Improve speed and reliability to load critical journey details (barcode, screen, session time, etc.) by securely encoding data into the URL
- Lumos Kiosk: Trailer video caching at a site-level

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- Lumos: Integrate Subs Portal & LT into Lumos Web as features, deprecate standalone sites and transition existing customers

Phase 2

- Captcha as part of Ticketing Flow.
- Lumos Web: SEO metadata tags and social media link sharing support
- Lumos: Use Licensing to restrict features
- LT: Reduce component duplication between OCC and LT (need to consider ODEON, aka zombie ticket, support)
- LT: Reconsider CloudFlare Worker usage for loading Journeys
- Porting to .Net API with simple CF Proxy would simplify logging support and simplify tooling. Alternatively, go all in and consider using CF R2 for Journey data...
- Lumos Mobile: Ensure an efficient process exists for publishing/deploying/updating a large number of applications (consider Ionic AppFlow)
- Omnia: Standardise deployment and monitoring to reduce cost to serve and be consistent with standard Vista Cloud processes

Phase 3

- Connect/Web Loader: Consider replacing with a horizon producer
- Connect/OCAPI: Consider replacing VISTAIT reference data with Horizon DB
- Connect/OCAPI: Consider Connect simply proxy to other services, e.g.: an Order Basket Service and keep a local cache of the orders/loyalty login sessions and providing auth
- Lumos CMS: Consider automating DNS provisioning for Lumos vanity domains as a self-service for customers via the CMS
- Lumos: Integrate Lighthouse tests into build process (perf, accessibility, seo, best practices)
- Lumos: Integrate OWASP ZAP security test tool into build process
- Omnia: Consider including Omnia front-ends in the Lumos monorepo

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Additional Security Elements

A couple of other elements that are not well homed in the Architecture roadmap, that will nonetheless impact our Platform engineering and workloads over the next few years.

SOC-2 Certification

Work required to bring Vista Cloud to be SOC-2 certified.

PCI/PA-DSS Certification

Work Required outside of the above scope towards PCI & PA-DSS Certification (now SSC)

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